

EC321 · MID-SEM

THE PAPERS

Three mid-sem papers — 2023, 2024 and 2025 — read end to end, every question worked out, and then a **predicted paper for September 2026** built from what actually repeats.

WHAT ACTUALLY GETS ASKED

3 PAPERS READ END TO END

Three papers is a small sample, but this department is unusually consistent: the same five or six things come back almost every year, in almost the same words. The rows tinted below appeared in **every single paper we have** — if you learn nothing else, learn those.

TOPIC	23	24	25	OF 3
ALOHA — draw the flow chart and explain it	■	■	■	3
Layers — name them and give each layer's job	■	■	■	3
PDR, throughput, latency and its components	■	■	■	3
Go-Back-N with timing and sliding-window diagrams	■	■	■	2
CDMA — build a Walsh table and decode	■	■	■	2
CSMA family (persistence methods, CD versus CA)	■	■	■	2
Topologies compared	■	■	■	2
Pure ALOHA throughput numerical	■	■	■	1
CSMA/CD minimum frame size numerical	■	■	■	1
Bandwidth-delay product and link utilisation	■	■	■	1
Connecting devices compared (hub, switch, router, gateway)	■	■	■	1
Selective Repeat ARQ, drawn step by step	■	■	■	1

TOPIC	23	24	25	OF 3
HDLC frame — read the fields off a diagram				1
Optical fibre structure				1
Five components of data communication				1
Physical versus IP versus port address				1

FILLED = IT APPEARED THAT YEAR · TINTED ROWS CAME UP IN EVERY PAPER WE HAVE

READ IT THIS WAY

Three things are effectively guaranteed. The ALOHA flow chart, the layer list, and the latency breakdown have been on every paper we have, and they are worth about 9 marks between them. They are also pure recall — there is no excuse for dropping any of it.

Two things are near-certain. Go-Back-N drawn properly and the CDMA Walsh table have both appeared in the last two years, and the CDMA question is worth 5 marks on its own. Note the instruction that came with it in 2024 and again in 2025: *“only theoretical explanation is not sufficient — draw the appropriate codes, data, matrix, graph.”* Words alone score nothing there.

The numericals rotate. 2023 was the numerical-heavy paper (ALOHA throughput, CSMA/CD frame size, bandwidth-delay utilisation) and 2024 and 2025 were almost entirely descriptive. That is the main risk for 2026: two descriptive years in a row makes a numerical return more likely, not less, and the three in the 2023 paper are the three that exist.

What is on the syllabus but has not been asked yet: CRC and Hamming worked by hand, bit and byte stuffing, and the learning bridge with the loop problem. All four are on the mid-sem topic list your sir gave out, and none has appeared in three years. Do not skip them.

PREDICTED PAPER — SEPTEMBER 2026

OUR READING, WITH FULL SOLUTIONS

This is **our own prediction**, not a leaked paper and not an official anything. It is built by taking every topic that has appeared in the last three years, weighting it by how often, checking it against the 37-topic mid-sem list, and then writing a paper of the same shape and mark total as the real ones. Treat it as a mock. Every question here has a full solution attached.

S.V. National Institute of Technology, Surat

Department of Electronics Engineering

MID Semester Exam, September 2026 — *predicted*

B. Tech. (EC) V Semester

COMMUNICATION NETWORKS (EC321)

MAXIMUM MARKS: 30

DURATION: 1:30 HOURS

ALL QUESTIONS COMPULSORY

Instructions. [CO] indicates course outcome. Marks are in the right-hand column. Where a diagram is asked for, a written description alone will not score.

Q.1 a What are the five components of a data communication system? State the role of each, and say which one is not physically located between the sender and the receiver. CO1

2

SOLUTION

The five components are **message, sender, receiver, transmission medium and protocol**.

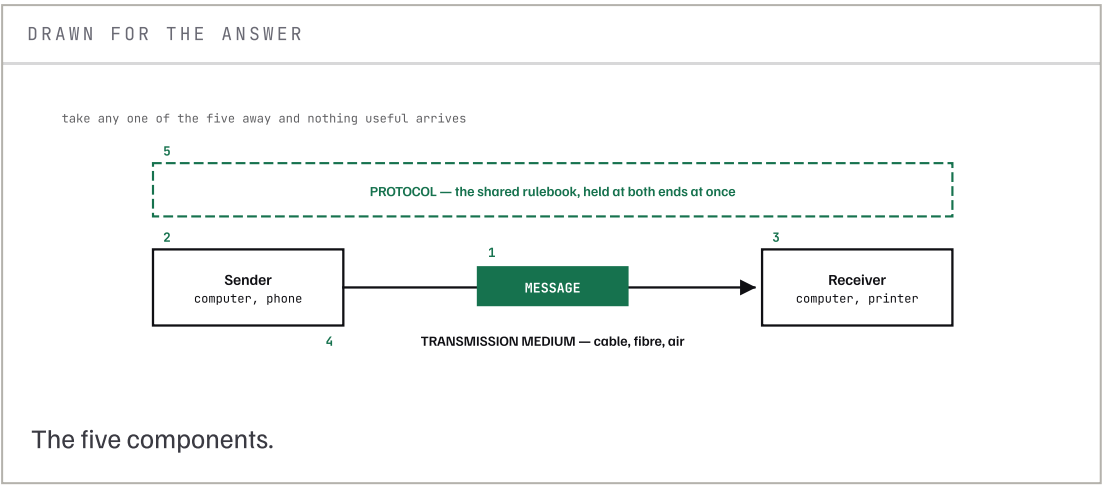
COMPONENT	ITS ROLE
Message	The information being communicated — text, numbers, pictures, audio, video.
Sender	The device that sends the message: a computer, phone, camera, sensor.
Receiver	The device that receives it.
Transmission medium	The physical path the message travels along: twisted pair, coax, fibre, or air.

COMPONENT	ITS ROLE
Protocol	The set of rules both ends agree on, so that the bits actually mean something at the far end.

The one that is **not** in the path is the **protocol**. It is not a wire or a box — it is an agreement held simultaneously at both ends.

THE LINE THAT EARNS THE MARK

Two devices can be joined by a perfect cable and still communicate nothing. A Hindi speaker and a Japanese speaker in the same room have message, sender, receiver and medium all present, and still exchange nothing. The missing piece is the protocol.



Q.1 b Compare the star, bus, ring and mesh topologies in terms of cost, robustness and performance. For a full mesh of n nodes, how many links are needed and how many ports must each node have? C01

3

SOLUTION

TOPOLOGY	LINKS	IF ONE LINK FAILS	MAIN WEAKNESS
Mesh	$n(n-1)/2$	Nothing else is affected — there is always another path	Cost. Cabling and port count explode with n
Star	n	Only that one station is cut off	The hub is a single point of failure
Bus	1 backbone + n drops	A break in the backbone can split or kill the network	Hard to fault-find; limited length

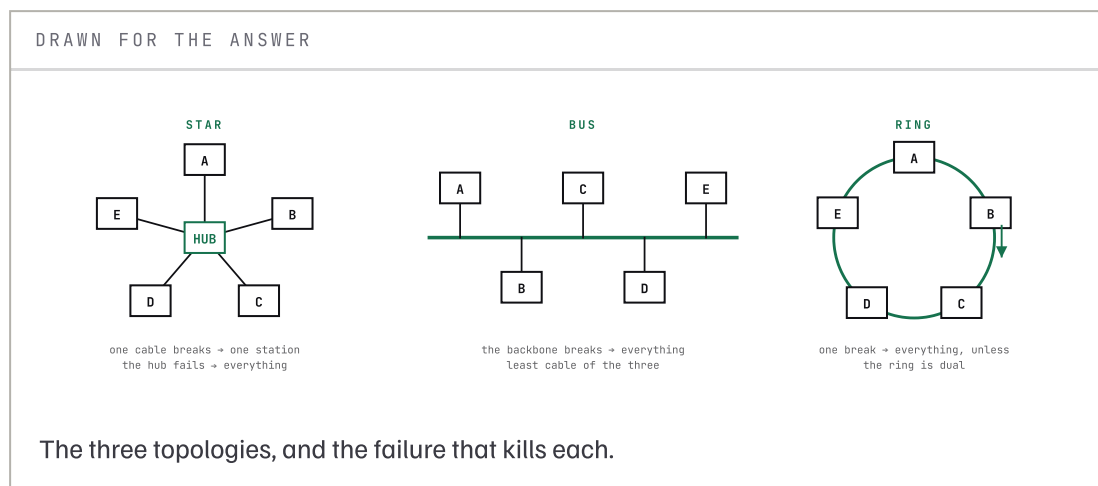
TOPOLOGY	LINKS	IF ONE LINK FAILS	MAIN WEAKNESS
Ring	n	A single break stops the ring unless it is dual	A signal passes through every station, so delay grows with n

Full mesh of n nodes:

number of links = $n(n - 1) / 2$
 ports per node = $n - 1$

Each node connects directly to every other node, which is $n-1$ links from each node. Multiplying $n \times (n-1)$ counts every link twice — once from each end — so divide by two.

For six nodes: **15 links, 5 ports each**



Q.1 c Draw the flow chart of the pure ALOHA protocol and explain each step.

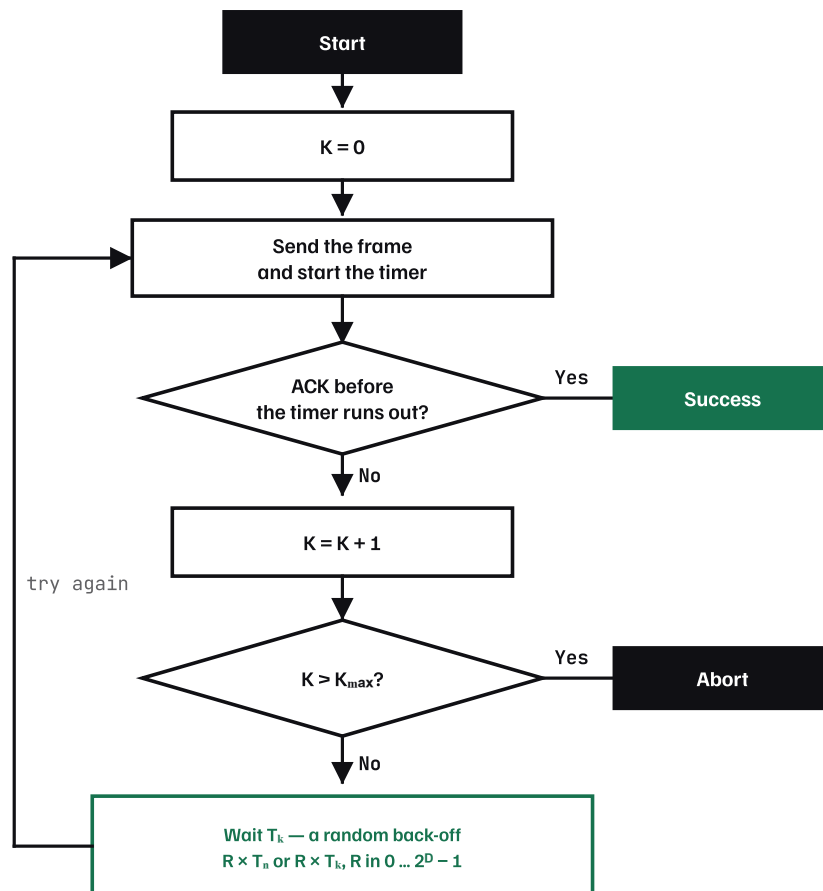
C02

3

SOLUTION

This question has been on every paper we have. Draw the chart — do not just describe it.

DRAWN FOR THE ANSWER



K counts the attempts on the current frame. The wait is random on purpose — if every station waited the same time they would all collide again together.

The pure ALOHA procedure. K counts the attempts on the current frame, and the back-off is random on purpose — if every station waited the same time they would all collide again together.

Step by step:

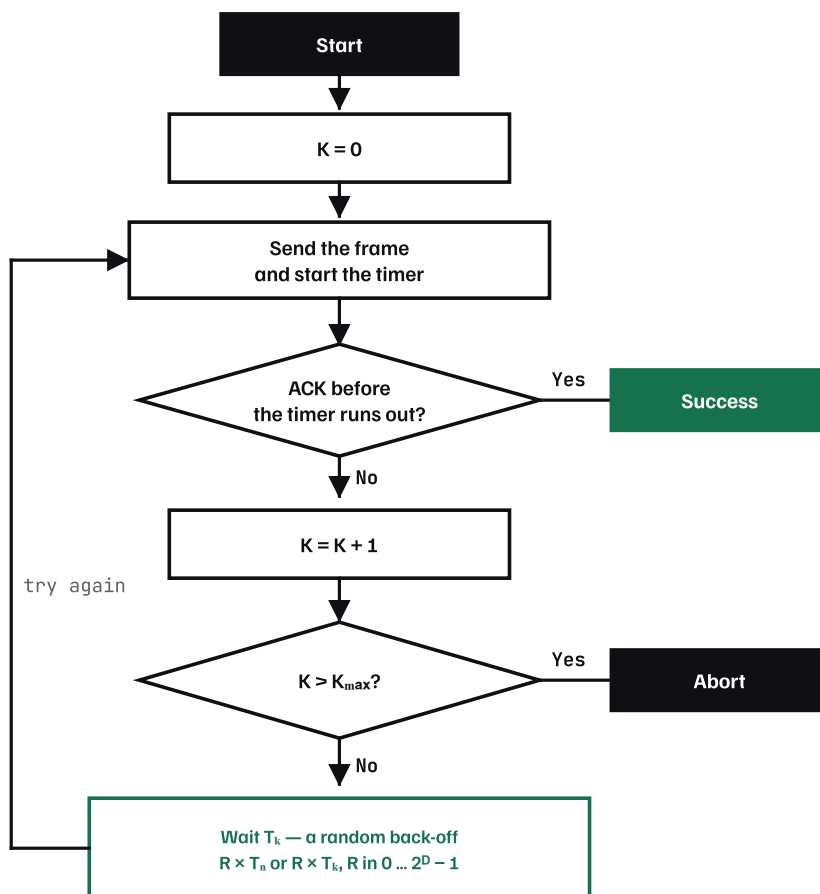
1. **Send immediately.** A pure ALOHA station transmits whenever it has a frame. It does not listen first — that is what separates ALOHA from CSMA.
2. **Wait one round trip.** The station waits $2 \times T_p$ for an acknowledgement, which is the longest a reply can take.
3. **Acknowledged?** If yes the frame got through and the station stops.
4. **Increment K.** If no acknowledgement came, a collision is assumed and the attempt counter goes up.
5. **Check the limit.** If K exceeds $K_{\max}$ (usually 15) the frame is abandoned — otherwise a permanently jammed channel would retry forever.

6. **Binary exponential back-off.** Pick R at random from 0 to $2^K - 1$ and wait $R \times T_p$ before retrying.

WHY THE BACK-OFF IS RANDOM AND DOUBLING

Random, because two stations that collided would otherwise retry at exactly the same instant and collide again forever. **Doubling**, because a repeated collision is evidence that the channel is busy, so each failure widens the window a station picks from and thins the traffic out.

DRAWN FOR THE ANSWER



K counts the attempts on the current frame. The wait is random on purpose – if every station waited the same time they would all collide again together.

The pure ALOHA flow chart.

Q.1 d Explain Packet Delivery Ratio, Throughput, Latency and Jitter. Break latency into its components and say what each one depends on. C01

3

SOLUTION

Packet Delivery Ratio — the fraction of sent packets that actually arrive, as a percentage. It measures **reliability**; a low PDR points to loss, congestion or a poor link.

$$\text{PDR} = (\text{packets received} / \text{packets sent}) \times 100\%$$

Throughput — the rate at which data is *successfully* transmitted, in bits per second.

$$\text{Throughput} = \text{total data successfully received} / \text{total time}$$

Throughput is not bandwidth. Bandwidth is the theoretical maximum the link could carry; throughput is what survives collisions, retransmissions, headers and congestion. It is always lower, often far lower.

Latency — the time for a packet to get from source to destination. It has **four** components:

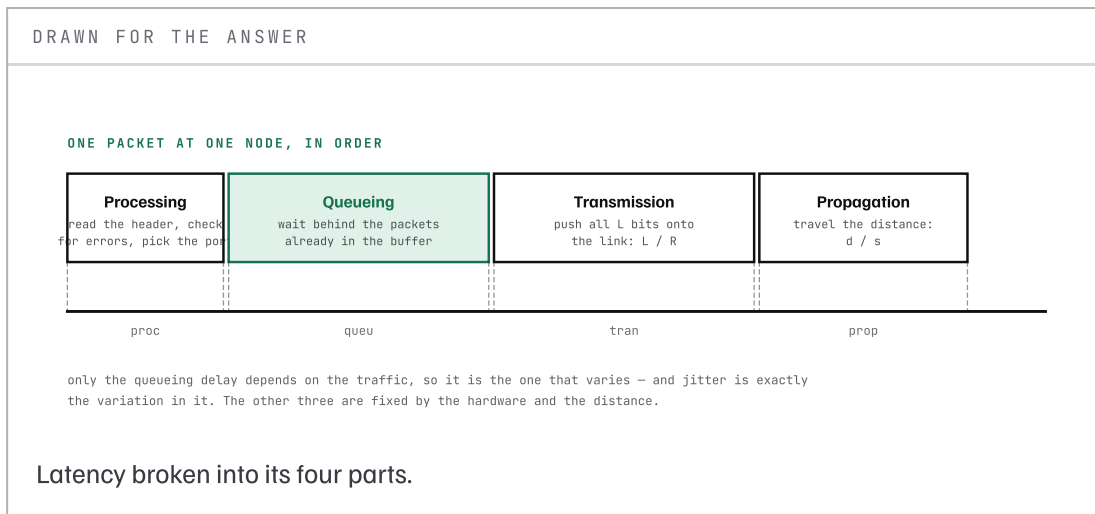
COMPONENT	WHAT IT IS	WHAT IT DEPENDS ON
Propagation delay	Time for the signal to travel the distance	distance ÷ propagation speed. Nothing else — not bandwidth, not frame size
Transmission delay	Time to push all the bits onto the link	frame size ÷ bandwidth
Processing delay	Time for routers and switches to handle the headers	device speed, number of hops
Queuing delay	Time spent waiting in buffers	traffic load — the only one that varies wildly

$$\text{Latency} = \text{propagation} + \text{transmission} + \text{processing} + \text{queuing}$$

Jitter — the variation in delay between consecutive packets.

$$\text{Jitter} = | D(i+1) - D(i) |$$

It is an absolute value, so a delay that *falls* still counts. High jitter ruins voice and video even when the average latency is fine, because packets arrive unevenly.



Q.1 e How does CSMA/CD work in Ethernet? Why can the same collision-detection method not be used in a wireless network, and what is used instead? C02

3

SOLUTION

CSMA/CD — Carrier Sense Multiple Access with Collision Detection.

- 1. Carrier sense.** Before transmitting, the station listens to the medium. If it is busy, it waits (1-persistent in Ethernet: it keeps listening and sends the moment it goes idle).
- 2. Transmit and keep listening.** While sending, the station monitors the line. If what it hears differs from what it is sending, a collision has occurred.
- 3. Abort and jam.** It stops immediately and sends a short jamming signal so that every other station also learns of the collision.
- 4. Back off.** It waits a random binary-exponential time and goes back to step 1.

For this to work the station must still be transmitting when the collision signal reaches it, which gives the minimum frame size condition:

$$T(\text{frame}) \geq 2 \times T(\text{propagation})$$

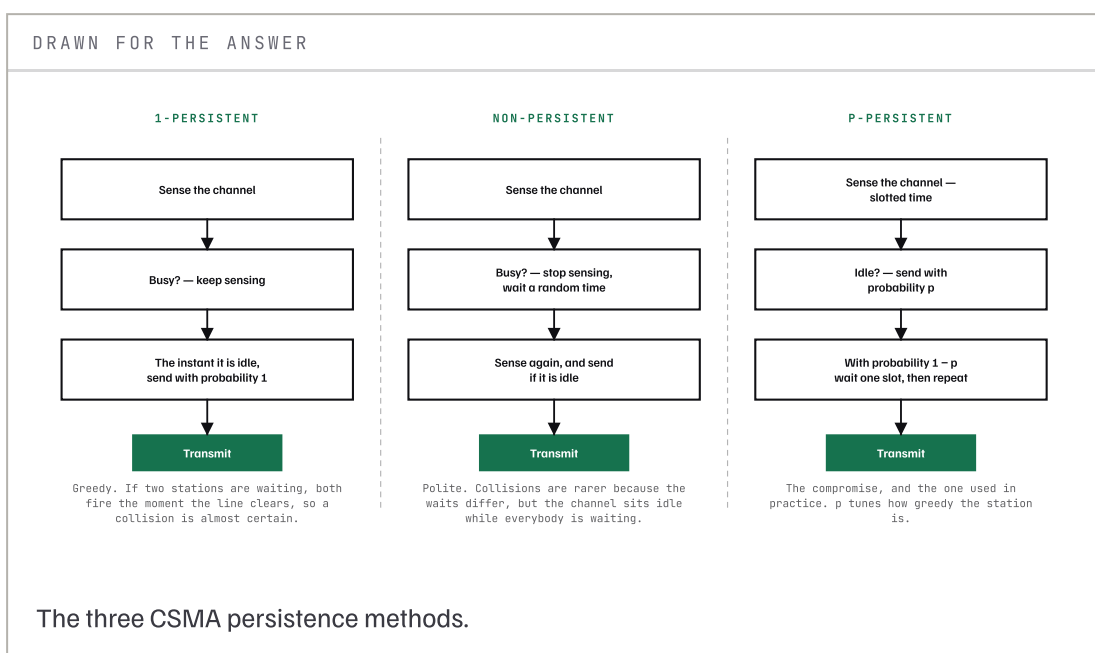
Why wireless cannot do it. Three separate reasons, and any of them alone is fatal:

- **The transmitter is deaf while it transmits.** Its own signal at its own antenna is many orders of magnitude stronger than anything arriving from a distant station, so it simply cannot hear a collision happening.

- **The hidden terminal problem.** A and C may both be in range of B but out of range of each other. Each senses an idle channel, both transmit, and the frames collide at B — and neither sender ever knows.
- **The exposed terminal problem.** The reverse case, where a station holds back unnecessarily because it hears a transmission that would not actually have interfered.

What is used instead: CSMA/CA — collision *avoidance*. Since a collision cannot be detected, it must be prevented:

- **Interframe space (IFS).** After the channel goes idle, wait a fixed gap before even starting the countdown. A shorter IFS means higher priority.
- **Contention window with back-off.** Choose a random slot count and count down only while the channel is idle, freezing the counter when it goes busy.
- **Explicit acknowledgement.** Silence cannot be read as success, so every frame must be acknowledged; no ACK means retransmit.
- **Optional RTS/CTS handshake** to solve the hidden terminal problem.



Q.1 f Differentiate between a physical address, an IP address and a port address. Explain what changes and what stays the same as a packet travels from a host on one network to a host on another. C01

3

SOLUTION

ADDRESS	FORM	SCOPE OF DELIVERY	USED BY
Physical (MAC)	48 bits, burned into the NIC, six hex octets	Node to node, inside one LAN only	Switch / bridge, data link layer
Logical (IP)	Assigned by software, hierarchical	Host to host, across networks	Router, network layer
Port	A 16-bit number identifying a process	Process to process	Transport layer

What changes on the way. This is the part worth the marks:

- The **IP addresses stay the same** from end to end. They identify the two hosts, and that does not change just because the packet crosses a router.
- The **MAC addresses change on every hop**. A frame is addressed to the next device on this LAN — often a router, not the final destination — so the router strips the old frame, keeps the packet, and builds a new frame with a new source and destination MAC for the next link.
- The **port numbers stay the same** throughout; they are only read at the two ends.

THE SENTENCE TO WRITE

A MAC address is flat — it carries no network part — so a router cannot use it to decide where to send something. That is exactly why the IP address exists, and why a frame addressed only by MAC can never leave its LAN.

Q.2 a Explain the concept of layering in a data communication network, and describe the function of each layer of the OSI model. Also state how many layers the TCP/IP suite uses. C02

5

SOLUTION

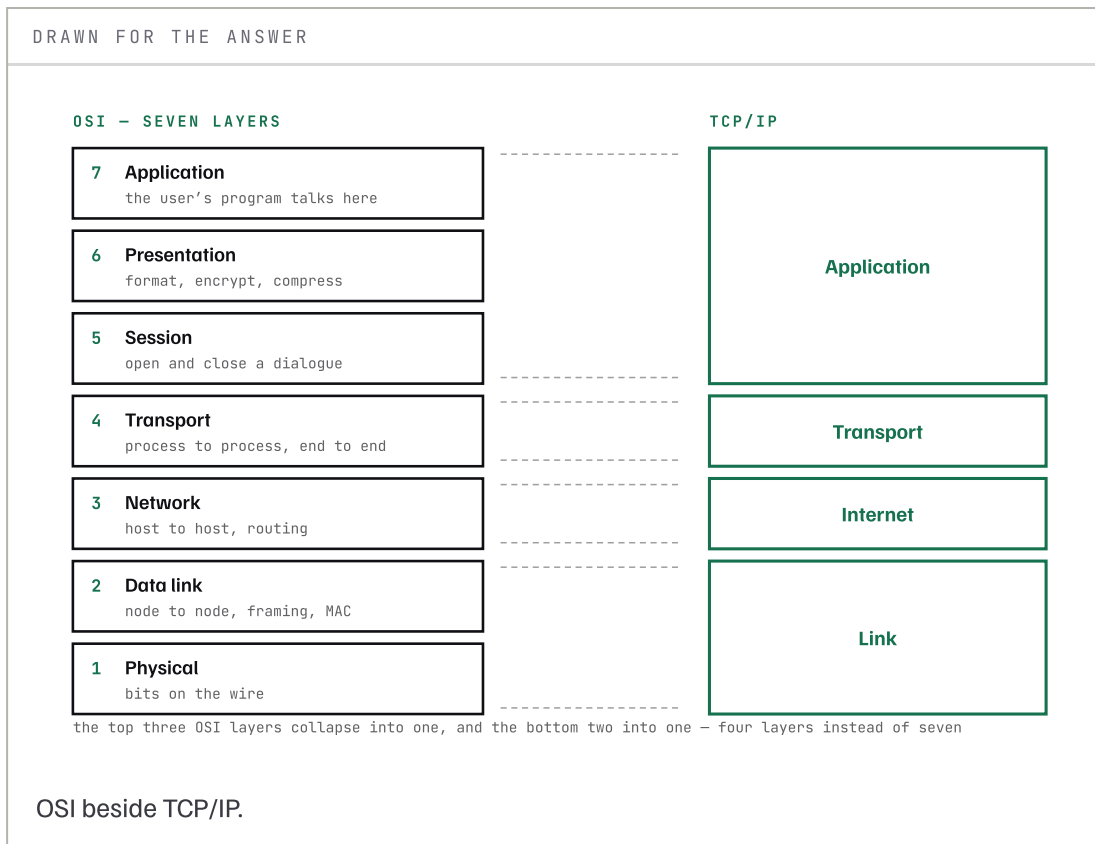
Why layers. A network is far too complicated to design as one lump. Layering cuts the job into pieces, gives each piece one clear duty, and lets each layer use the one below without knowing how it works. The benefits: a layer can be replaced without touching the others, different vendors can implement different layers, and faults can be isolated to one layer.

The seven OSI layers, bottom to top:

#	LAYER	ITS JOB	UNIT
1	Physical	Transmits raw bits over the medium: voltages, timing, connectors, data rate	bits
2	Data link	Node-to-node delivery: framing, physical addressing, flow control, error control, access control	frame
3	Network	Source-to-destination delivery across networks: logical addressing and routing	packet
4	Transport	Process-to-process delivery: segmentation and reassembly, port addressing, end-to-end flow and error control	segment
5	Session	Establishes, manages and synchronises the dialogue between the two ends	data
6	Presentation	Translation, encryption and compression — the syntax and semantics of the data	data
7	Application	Services to the user: mail, file transfer, web access	data

Encapsulation. Going down the stack each layer adds its own header, wrapping whatever came from above as pure data. Going up at the receiver each layer strips its own header — decapsulation. That is why a layer never has to understand the layers above it.

TCP/IP uses **five** layers in the practical drawing: physical, data link, network, transport and application. It collapses OSI's session, presentation and application into the single application layer.



Q.2 b In CDMA, design a Walsh table for a network of four nodes A, B, C and D (2 marks). Assign one code to each node. Node A transmits bit '1', node B transmits bit '0', node C transmits bit '1', and node D is receiving. Show how D can detect the bits transmitted by A, B and C (3 marks). **A theoretical explanation alone is not sufficient — draw the codes, the data and the arithmetic.** C01

5

SOLUTION

Step 1 — build the Walsh table. Walsh matrices double in size by the rule

DRAWN FOR THE ANSWER

WALSH W_4

A	+1	+1	+1	+1
B	+1	-1	+1	-1
C	+1	+1	-1	-1
D	+1	-1	-1	+1

built from W_2 by $[W \ W; \ W \ -W]$

the codes are orthogonal, so every other station's contribution cancels to zero

A SENDS 1 · B SENDS 0 · C SENDS 1 · D LISTENS

$$A \times (+1) \quad \quad \quad +1 \ +1 \ +1 \ +1$$

$$B \times (-1) \quad \quad \quad -1 \ +1 \ -1 \ +1$$

$$C \times (+1) \quad \quad \quad +1 \ +1 \ -1 \ -1$$

$$\text{sum on the air} \quad \quad \quad +1 \ +3 \ -1 \ +1$$

D wants A's bit, so D multiplies by A's code and averages:

$$\begin{aligned} & (+1)(+1) + (+3)(+1) + (-1)(+1) + (+1)(+1) = 4 \\ & 4 / 4 = +1 \rightarrow \text{A sent a 1} \end{aligned}$$

The Walsh table, and one decode carried all the way through: D multiplies the sum on the air by A's own code, averages, and gets A's bit back because every other code is orthogonal to it.

$$\begin{array}{lcl} W(2) = \begin{bmatrix} +1 & +1 \\ +1 & -1 \end{bmatrix} & & W(4) = \begin{bmatrix} +1 & +1 & +1 & +1 \\ +1 & -1 & +1 & -1 \\ +1 & +1 & -1 & -1 \\ +1 & -1 & -1 & +1 \end{bmatrix} \rightarrow \begin{array}{l} A \\ B \\ C \\ D \end{array} \end{array}$$

Every pair of rows is **orthogonal** — multiply any two different rows element by element and the four products add to zero. That property is the whole trick.

Step 2 — what each station puts on the channel. A 1 is sent as $(+1 \times \text{its code})$, a 0 as $(-1 \times \text{its code})$, and a silent station contributes nothing.

$$\begin{aligned} \text{A sends 1 : } & +1 \times [+1 \ +1 \ +1 \ +1] = [+1 \ +1 \ +1 \ +1] \\ \text{B sends 0 : } & -1 \times [+1 \ -1 \ +1 \ -1] = [-1 \ +1 \ -1 \ +1] \\ \text{C sends 1 : } & +1 \times [+1 \ +1 \ -1 \ -1] = [+1 \ +1 \ -1 \ -1] \\ \text{D is receiving, so it sends nothing.} & \end{aligned}$$

$$\text{channel (add column by column)} = \begin{bmatrix} +1 & +3 & -1 & +1 \end{bmatrix}$$

Step 3 — D decodes each station in turn. Multiply the channel signal element by element by that station's code, add the four products, and divide by $N = 4$.

$$\begin{aligned} \text{For A: } & [+1 \ +3 \ -1 \ +1] \cdot [+1 \ +1 \ +1 \ +1] \\ & = (+1) + (+3) + (-1) + (+1) = +4 \rightarrow +4/4 = +1 \rightarrow \text{bit 1} \end{aligned}$$

$$\begin{aligned} \text{For B: } & [+1 \ +3 \ -1 \ +1] \cdot [+1 \ -1 \ +1 \ -1] \\ & = (+1) + (-3) + (-1) + (-1) = -4 \rightarrow -4/4 = -1 \rightarrow \text{bit 0} \end{aligned}$$

$$\begin{aligned} \text{For C: } & [+1 \ +3 \ -1 \ +1] \cdot [+1 \ +1 \ -1 \ -1] \\ & = (+1) + (+3) + (+1) + (-1) = +4 \rightarrow +4/4 = +1 \rightarrow \text{bit 1} \end{aligned}$$

D recovers A = 1, B = 0, C = 1

WHY IT WORKS, IN ONE LINE

Each station's own code multiplied by itself gives +1 in every position, so its contribution adds up to $\pm N$. Every *other* station's code is orthogonal to it, so those contributions cancel to zero. Dividing by N leaves exactly ± 1 — the wanted bit, with everybody else subtracted out.

If the question asks for six nodes, you cannot use $W(4)$. Build $W(8)$ the same way and use any six of its eight rows — Walsh tables only exist for sizes that are powers of two.

DRAWN FOR THE ANSWER

WALSH W_4

A	+1	+1	+1	+1
B	+1	-1	+1	-1
C	+1	+1	-1	-1
D	+1	-1	-1	+1

A SENDS 1 · B SENDS 0 · C SENDS 1 · D LISTENS

A × (+1) +1 +1 +1 +1

B × (-1) -1 +1 -1 +1

C × (+1) +1 +1 -1 -1

sum on the air +1 +3 -1 +1

D wants A's bit, so D multiplies by A's code and averages:

$$\begin{aligned} & (+1)(+1) + (+3)(+1) + (-1)(+1) + (+1)(+1) = 4 \\ & 4 / 4 = +1 \rightarrow \text{A sent a 1} \end{aligned}$$

built from W_2 by $\begin{bmatrix} W & W \\ W & -W \end{bmatrix}$

the codes are orthogonal, so every other station's contribution cancels to zero

The Walsh table and a worked decode.

Q.3 a A pure ALOHA network transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system produces 1000 frames per second? What would it be if slotted ALOHA were used instead? C01

3

SOLUTION

Step 1 — frame time.

$$T(\text{fr}) = \text{frame bits} / \text{bit rate} = 200 / 200,000 = 1 \text{ ms}$$

Step 2 — offered load G. G is the average number of frames generated during one frame time.

$$G = 1000 \text{ frames/s} \times 1 \text{ ms} = 1$$

Step 3 — pure ALOHA throughput.

$$S = G e^{-2G} = 1 \times e^{-2} = 0.135$$

So only 13.5% of the channel's frame slots carry a successful frame. The channel can hold 1000 frames per second, so

about 135 frames per second get through

Step 4 — slotted ALOHA on the same load.

$$S = G e^{-G} = 1 \times e^{-1} = 0.368 \rightarrow \text{about 368 frames per second}$$

THE COMPARISON THAT EARNS THE LAST MARK

Slotting forces every station to begin only at a slot boundary. That halves the vulnerable time from $2 \cdot T_{fr}$ to T_{fr} , and halving the window is exactly what doubles the peak throughput — from **18.4% at $G = 0.5$** for pure ALOHA to **36.8% at $G = 1$** for slotted.

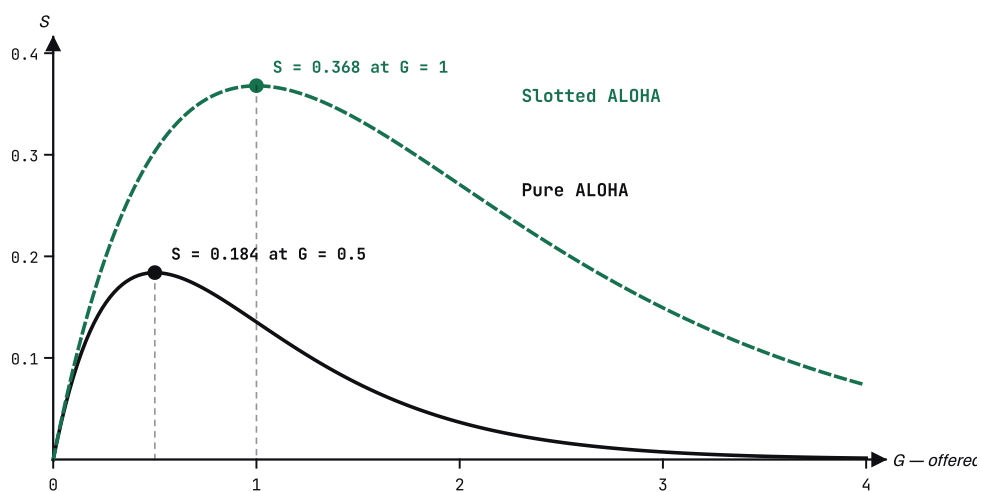
COMPUTED · TOOLS/MAKE-FIGURES.PY



Computed figure

Where the two peaks sit, and why offering more load past them delivers less.

DRAWN FOR THE ANSWER



past the peak, offering more traffic delivers less — the extra load is all collision

Where the two ALOHA peaks sit, and why past them more traffic means less throughput.

- Q.3 b** A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time is 25.6 μs , what is the minimum frame size? Explain why such a minimum has to exist at all. C01

2

SOLUTION

The condition. A station must still be transmitting when the worst-case collision signal gets back to it — otherwise it would finish, believe it succeeded, and never learn that the frame was destroyed.

$$T(\text{frame}) \geq 2 \times T(\text{propagation}) = 2 \times 25.6 \text{ us} = 51.2 \text{ us}$$

Convert time to bits.

$$\begin{aligned} \text{minimum frame} &= \text{bit rate} \times T(\text{frame}) \\ &= 10 \times 10^6 \times 51.2 \times 10^{-6} \\ &= 512 \text{ bits} \end{aligned}$$

$$512 \text{ bits} = 64 \text{ bytes}$$

This is exactly the 64-byte minimum frame size of standard Ethernet, and it is where that number comes from.

WATCH THE WORDING

If the question gives the **round-trip** time rather than the one-way propagation time, do not double it again. The 2023 paper gave $T_p = 51.2 \mu\text{s}$ one-way, which doubles to 102.4 μs and gives 1024 bits — 128 bytes, not 64. Read which one you have been given.

- Q.3 c** In a Stop-and-Wait ARQ system the bandwidth of the line is 1 Mbps and one bit takes 20 ms to make a round trip. Find the bandwidth-delay product. If the data frames are 1000 bits long, what is the utilisation of the link? What window size would be needed to reach 100%? C01

3

SOLUTION

Step 1 — bandwidth-delay product. This is how many bits the “pipe” holds.

```
BDP = bandwidth x round-trip time  
    = 1 x 10^6 x 20 x 10^-3  
    = 20,000 bits
```

Step 2 — Stop-and-Wait utilisation. Stop-and-Wait puts exactly one frame in flight per round trip.

```
utilisation = frame size / BDP = 1000 / 20,000 = 0.05
```

5% – the link is idle 95% of the time

Step 3 — the window needed for 100%.

```
window = BDP / frame size = 20,000 / 1000 = 20 frames
```

With a window of 15 the utilisation would be $15 \times 1000 / 20,000 = 75\%$. The rule to remember: **the window must be at least the bandwidth-delay product divided by the frame size.**

The real paper, exactly as set. Every question is worked below.

S.V. National Institute of Technology, Surat

Department of Electronics Engineering

MID Semester Exam, September 2025

B. Tech. (EC) V Semester · 24/09/2025, 11:15 AM

COMMUNICATION NETWORKS (EC321)

MAXIMUM MARKS: 30

DURATION: 1:30 HOURS

ALL QUESTIONS COMPULSORY

Q.1 a

What are the basic components of a data communication system, and what role does each component play?

C01

2

SOLUTION

See the predicted Q.1 a above — the same five components, the same table.
Message, sender, receiver, transmission medium, protocol. The protocol is the only one not physically in the path between the two ends.

DRAWN FOR THE ANSWER

take any one of the five away and nothing useful arrives

The five components. The protocol is the only one that is not physically between the two ends — it is an agreement held at both.

Q.1 b

Compare the star, bus and ring network topologies in terms of their advantages and disadvantages, with reference to performance parameters.

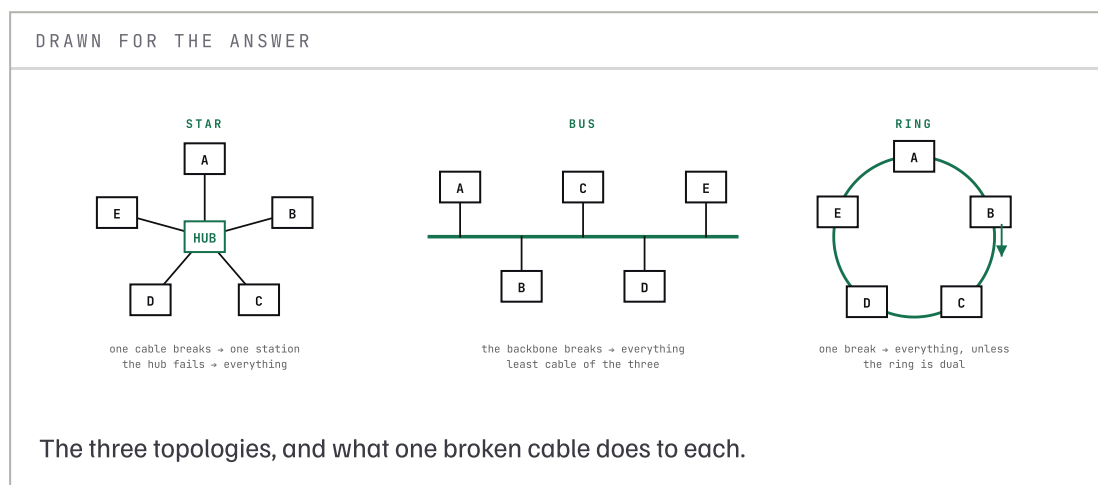
C01

2

SOLUTION

	STAR	BUS	RING
Cabling	n links, one per station	One backbone plus short drops — least cable	n links
Single failure	Isolates one station only	A backbone break can bring down the whole network	Breaks the ring unless it is dual
Weak point	The hub — if it dies, everything dies	The backbone, and reflections at a break	Any one station or link
Latency	Two hops through the hub, constant	Shared medium — grows with load through contention	Grows with the number of stations, since the signal passes through each
Throughput	Good; with a switch each pair gets full bandwidth	Falls as load rises because collisions rise	Predictable under token control
Adding a station	Easy — one new cable to the hub	Easy, but the backbone has a length limit	Requires breaking the ring

The mark is in the *performance* column, not the picture. The examiner is looking for the trade-off: the bus saves cable and pays in reliability and contention; the star spends cable and buys isolation; the ring gives predictable access and pays in delay.



Q.1 c Draw the flow chart of the ALOHA protocol and explain it. C02

3

SOLUTION

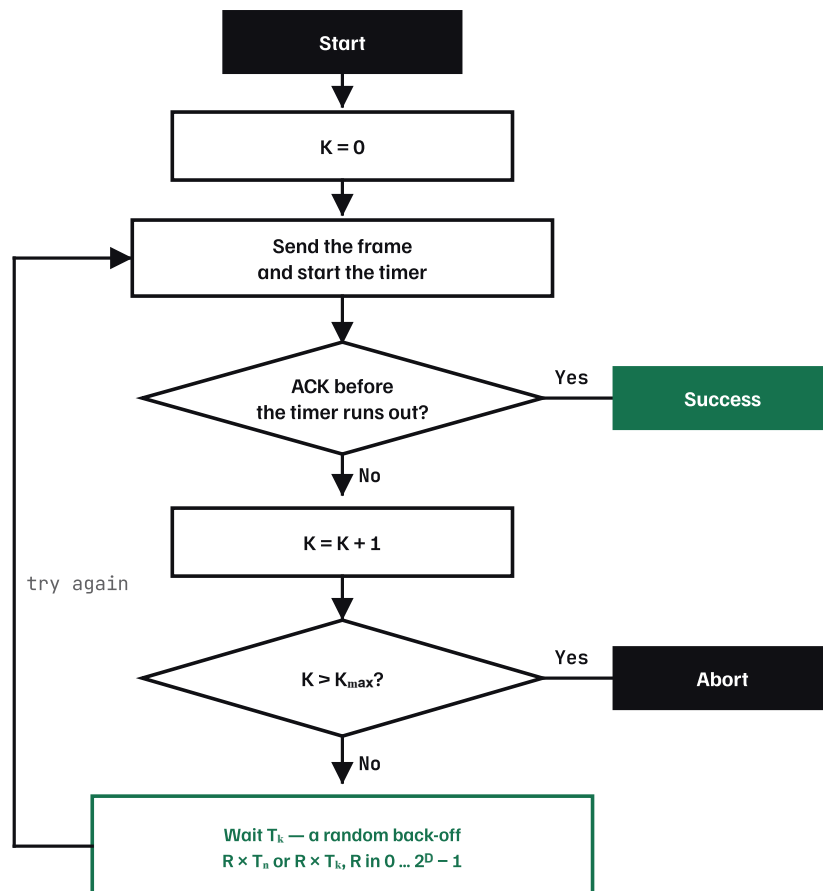
Identical to the predicted Q.1 c above: the flow chart with K , the $2 \times T_p$ wait, the $K_{\max}$ check and the binary exponential back-off, plus the explanation of why the back-off must be both random and doubling. **This exact question has now appeared three years running.**

COMPUTED · TOOLS/MAKE-FIGURES.PY

**Computed figure**

40,000 slots per point with Poisson arrivals. The simulated dots land on Ge^{-2v} and Ge^{-v} without being told to, and the two peaks come out at 0.184 and 0.368.

DRAWN FOR THE ANSWER



K counts the attempts on the current frame. The wait is random on purpose – if every station waited the same time they would all collide again together.

The pure ALOHA procedure, step by step.

Q.1 d How does the CSMA/CD protocol work in Ethernet, and why can the same collision-detection method not be used in wireless networks, which use CSMA/CA instead? C04

3

SOLUTION

Worked in full at the predicted Q.1 e above: the four steps of CSMA/CD, the $T_{\text{frame}} \geq 2 \cdot T_{\text{prop}}$ condition, then the three reasons wireless cannot detect collisions (a transmitter is deaf to its own band, hidden terminals, exposed terminals) and the four things CSMA/CA does instead.

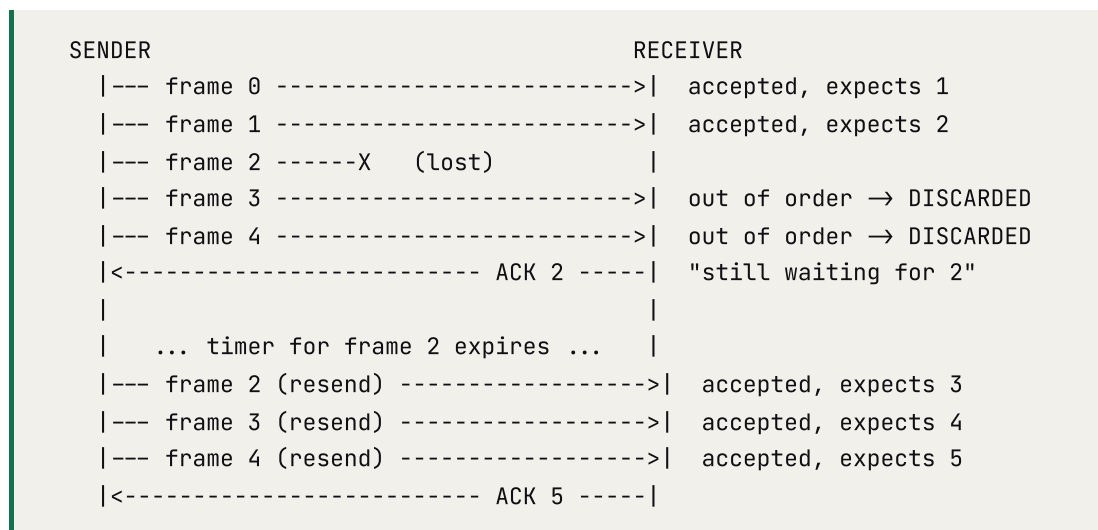
Q.1 e Explain the Go-back-N protocol with proper timing diagrams and sliding window diagrams. C02

3

SOLUTION

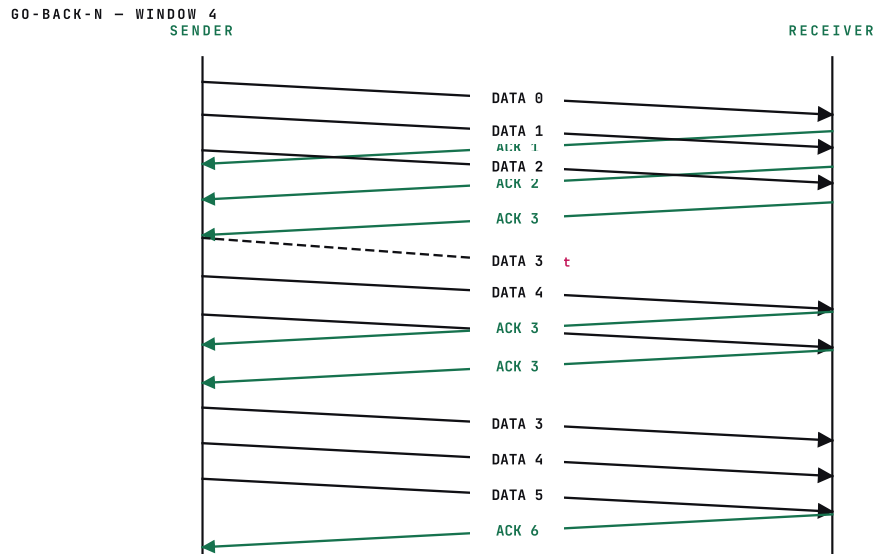
The idea. The sender may have up to $2^m - 1$ unacknowledged frames outstanding. The receiver window is **1**, so anything arriving out of order is discarded. On a timeout the sender goes back to the oldest unacknowledged frame and resends **everything** from there.

Timing diagram — frame 2 is lost, $m = 3$ so the window is 7:



Sliding window, step by step:

DRAWN FOR THE ANSWER



DATA 3 is lost. 4 and 5 do arrive, but the receiver window is 1, so it is still waiting for 3 and throws them away – each time repeating ACK 3. When the timer for 3 expires the sender goes back and resends 3, 4 and 5.

The same exchange as a timing diagram, which is what the question asks you to draw.

The window slides forward only when a cumulative acknowledgement arrives, and **ACK n** means “everything up to $n-1$ is safely received”, not “frame n arrived”.

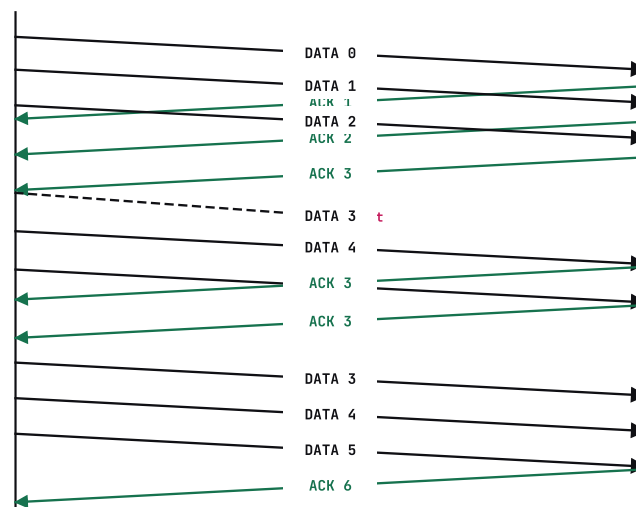
THE COMPARISON THE EXAMINER WANTS

Frames 3 and 4 arrived *perfectly* and were thrown away. That is the cost of Go-Back-N, and it is the whole reason Selective Repeat exists — SR buffers them and resends only frame 2, at the price of buffers and sorting at the receiver, and windows capped at 2^{m-1} instead of 2^m-1 .

DRAWN FOR THE ANSWER

GO-BACK-N - WINDOW 4
SENDER

RECEIVER



DATA 3 is lost. 4 and 5 do arrive, but the receiver window is 1, so it is still waiting for 3 and throws them away – each time repeating ACK 3. When the timer for 3 expires the sender goes back and resends 3, 4 and 5.

Go-Back-N: three frames acknowledged, one lost, and the whole window sent again.

Q.1 f Explain Packet delivery ratio, Throughput and Latency and its components. C01

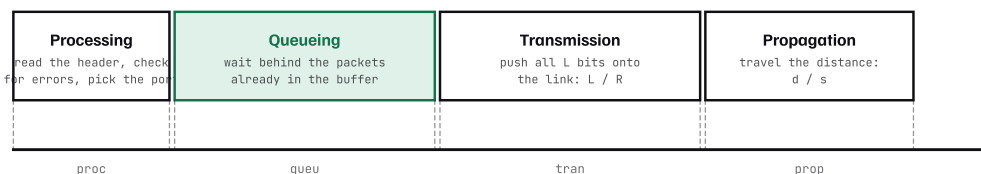
3

SOLUTION

Worked in full at the predicted Q.1 d above, including the jitter formula. The four latency components — propagation, transmission, processing, queuing — are the part that carries the marks, along with what each one depends on.

DRAWN FOR THE ANSWER

ONE PACKET AT ONE NODE, IN ORDER



only the queueing delay depends on the traffic, so it is the one that varies – and jitter is exactly the variation in it. The other three are fixed by the hardware and the distance.

Latency, broken into its four components. Only the queueing part varies with traffic, which is why it is the one that causes jitter.

Q.1 g Differentiate between a physical address, an IP address and a port address in the context of networking. What is each one used for? C01

4

SOLUTION

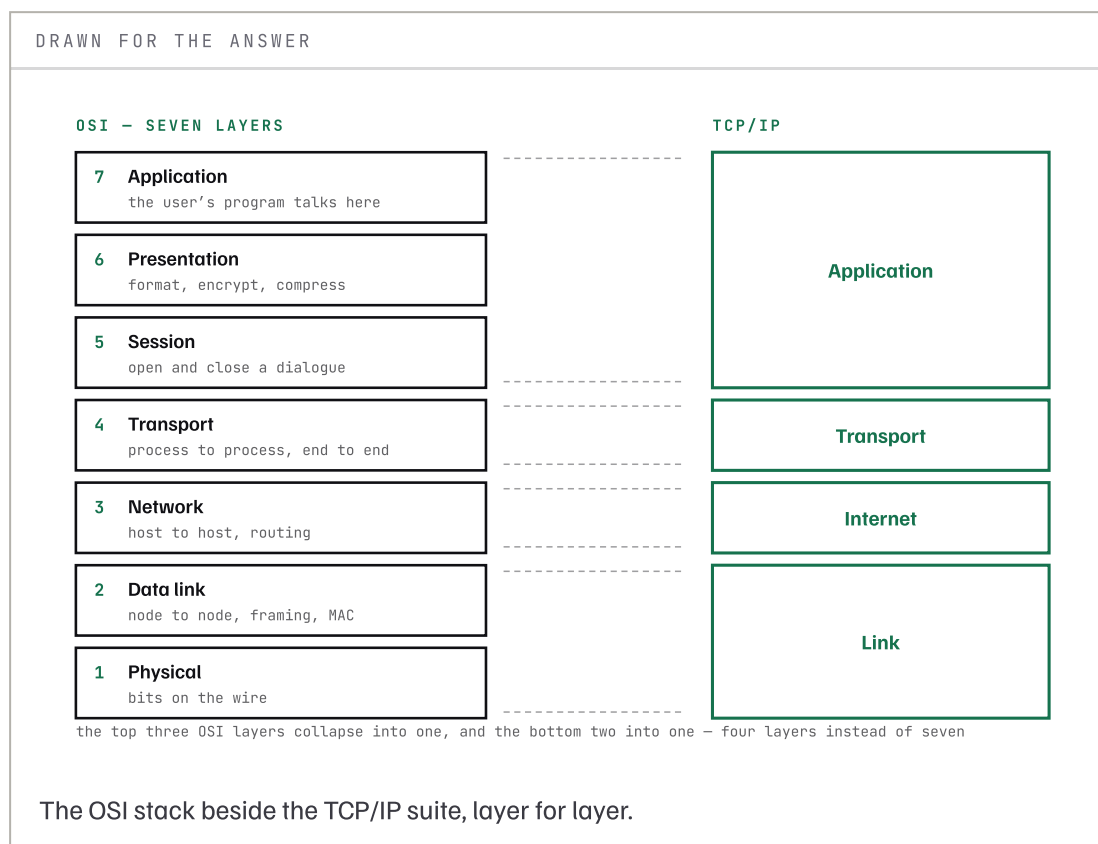
Worked in full at the predicted Q.1 f above, including the part that separates a good answer from a list: the IP addresses and port numbers stay the same end to end, while the MAC addresses are replaced on every single hop.

Q.2 a Explain the concept of layering in data communication networks and describe the functions of each layer of the OSI model. C02

5

SOLUTION

Worked in full at the predicted Q.2 a above: why layering exists, the seven-layer table with each layer's job and its data unit, encapsulation and decapsulation, and the note that TCP/IP uses five.



Q.2 b In case of CDMA, design a Walsh table for a network having 6 nodes (A, B, C, D, E, F) [2 marks]. Assign one code to each node. Node A is transmitting bit '1', B is transmitting bit '0' and C is transmitting bit '1', and node D is receiving. Show

5

how D can detect the bits transmitted by A, B and C [1+1+1 marks]. **Only theoretical explanation is not sufficient. Draw the appropriate codes / data / matrix / graph.** C01

SOLUTION

Six nodes needs W(8), not W(4) — Walsh matrices only exist for sizes that are powers of two, so take the next one up and use six of its eight rows. That single decision is the first mark.

W(8), rows assigned A..F (rows 7 and 8 unused):

```
A : +1 +1 +1 +1 +1 +1 +1 +1
B : +1 -1 +1 -1 +1 -1 +1 -1
C : +1 +1 -1 -1 +1 +1 -1 -1
D : +1 -1 -1 +1 +1 -1 -1 +1
E : +1 +1 +1 +1 -1 -1 -1 -1
F : +1 -1 +1 -1 -1 +1 -1 +1
```

What goes on the channel. A sends 1 (+1 × code), B sends 0 (−1 × code), C sends 1 (+1 × code); D, E and F are silent.

```
A (bit 1) : +1 +1 +1 +1 +1 +1 +1 +1
B (bit 0) : -1 +1 -1 +1 -1 +1 -1 +1
C (bit 1) : +1 +1 -1 -1 +1 +1 -1 -1
-----
channel   : +1 +3 -1 +1 +1 +3 -1 +1
```

D decodes each in turn — multiply element by element by that node's code, add the eight products, divide by 8.

```
A: (+1)(+1)+(+3)(+1)+(-1)(+1)+(+1)(+1)
   +(+1)(+1)+(+3)(+1)+(-1)(+1)+(+1)(+1)
   = 1+3-1+1+1+3-1+1 = +8 → +8/8 = +1 → bit 1
```

```
B: (+1)(+1)+(+3)(-1)+(-1)(+1)+(+1)(-1)
   +(+1)(+1)+(+3)(-1)+(-1)(+1)+(+1)(-1)
   = 1-3-1-1+1-3-1-1 = -8 → -8/8 = -1 → bit 0
```

```
C: (+1)(+1)+(+3)(+1)+(-1)(-1)+(+1)(-1)
   +(+1)(+1)+(+3)(+1)+(-1)(-1)+(+1)(-1)
   = 1+3+1-1+1+3+1-1 = +8 → +8/8 = +1 → bit 1
```

D recovers A = 1, B = 0, C = 1

Orthogonality is what makes this work: a code multiplied by itself gives +1 in every position and sums to +N, while any two different Walsh rows sum to exactly zero — so everybody else subtracts out cleanly.

DRAWN FOR THE ANSWER

WALSH W_4

A	+1	+1	+1	+1
B	+1	-1	+1	-1
C	+1	+1	-1	-1
D	+1	-1	-1	+1

A SENDS 1 · B SENDS 0 · C SENDS 1 · D LISTENS

A × (+1)

+1 +1 +1 +1

B × (-1)

-1 +1 -1 +1

C × (+1)

+1 +1 -1 -1

sum on the air

+1 +3 -1 +1

D wants A's bit, so D multiplies by A's code and averages:

(+1)(+1) + (+3)(+1) + (-1)(+1) + (+1)(+1) = 4

4 / 4 = +1 → A sent a 1

built from W_2 by [W W ; W -W]

the codes are orthogonal, so every other station's contribution cancels to zero

The Walsh table for four nodes, and D pulling A's bit out of the sum on the air.

18 SEPTEMBER 2026

28 / 48

MID-SEM, SEPTEMBER 2024

30 MARKS · 27 SEPT 2024

The scan of this paper is poor in places, so a couple of sub-questions are reconstructed from what is legible. Where that is the case it is said so.

S.V. National Institute of Technology, Surat

Department of Electronics Engineering

MID Semester Exam, September 2024

B. Tech. (EC) V Semester · 27/09/2024

COMMUNICATION NETWORKS

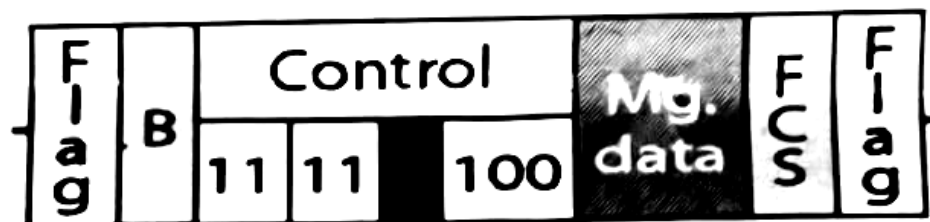
MAXIMUM MARKS: 30

DURATION: 1:30 HOURS

ALL QUESTIONS COMPULSORY

- Q.1 a** Answer the following about the frame shown in the figure: (i) which protocol does it belong to? (ii) what type of frame is it? (iii) in which phase is it used? (iv) name one method that can be used for the FCS. (v) what is B'? 5

FROM THE PAPER



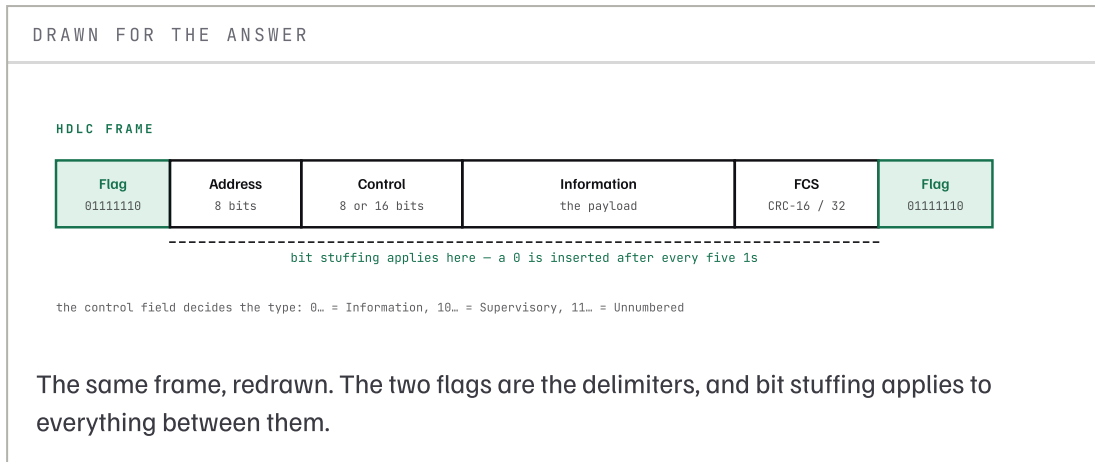
The frame as it is printed on the paper.

SOLUTION

The frame drawn is **Flag · Address · Control · Data · FCS · Flag** with the flag shown as 01111110, so:

- Which protocol:** HDLC — High-level Data Link Control. The 01111110 flag with bit stuffing and that field order is HDLC's signature.
- Type of frame:** read the first bits of the **control field**. First bit 0 → **I-frame** (carries data). 10 → **S-frame** (supervisory: acknowledgements and flow control, no data). 11 → **U-frame** (unnumbered: manages the link). A frame shown with a data field is an I-frame.
- Phase:** I-frames and S-frames belong to the **data transfer** phase; U-frames belong to the **connection establishment and release** phase.

4. **FCS method: CRC** — a cyclic redundancy check, normally CRC-16 or CRC-32. A checksum is the weaker alternative.
5. **What is B': the address field**, which in HDLC carries the address of the *secondary* station — because the primary is always one end of the link, naming the secondary is enough to identify the other end in both directions.



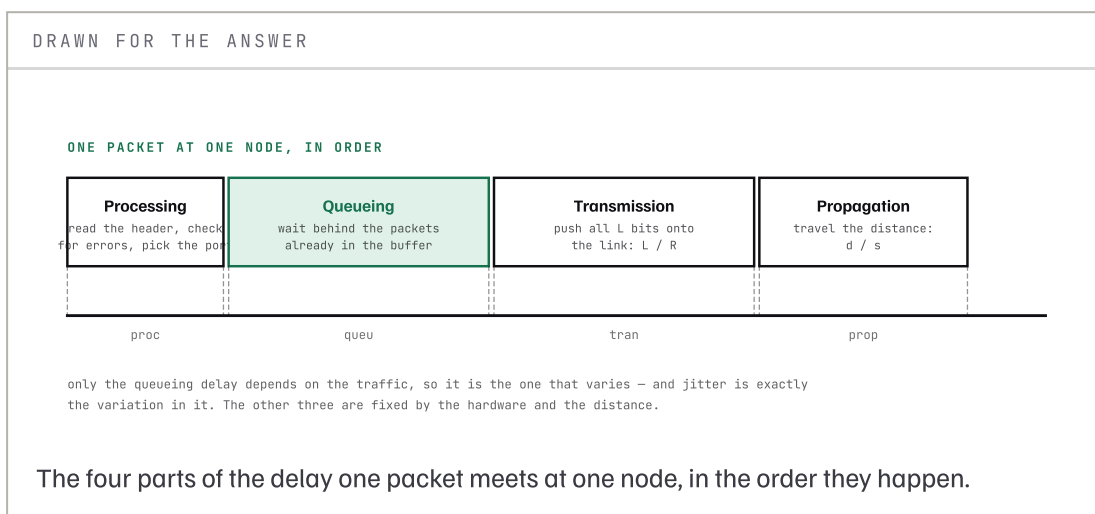
Q.1 b Explain latency and its components.

C01

3

SOLUTION

Same as the 2023 and 2025 papers — the four components with what each depends on, and the total. Worked at the predicted Q.1 d above.



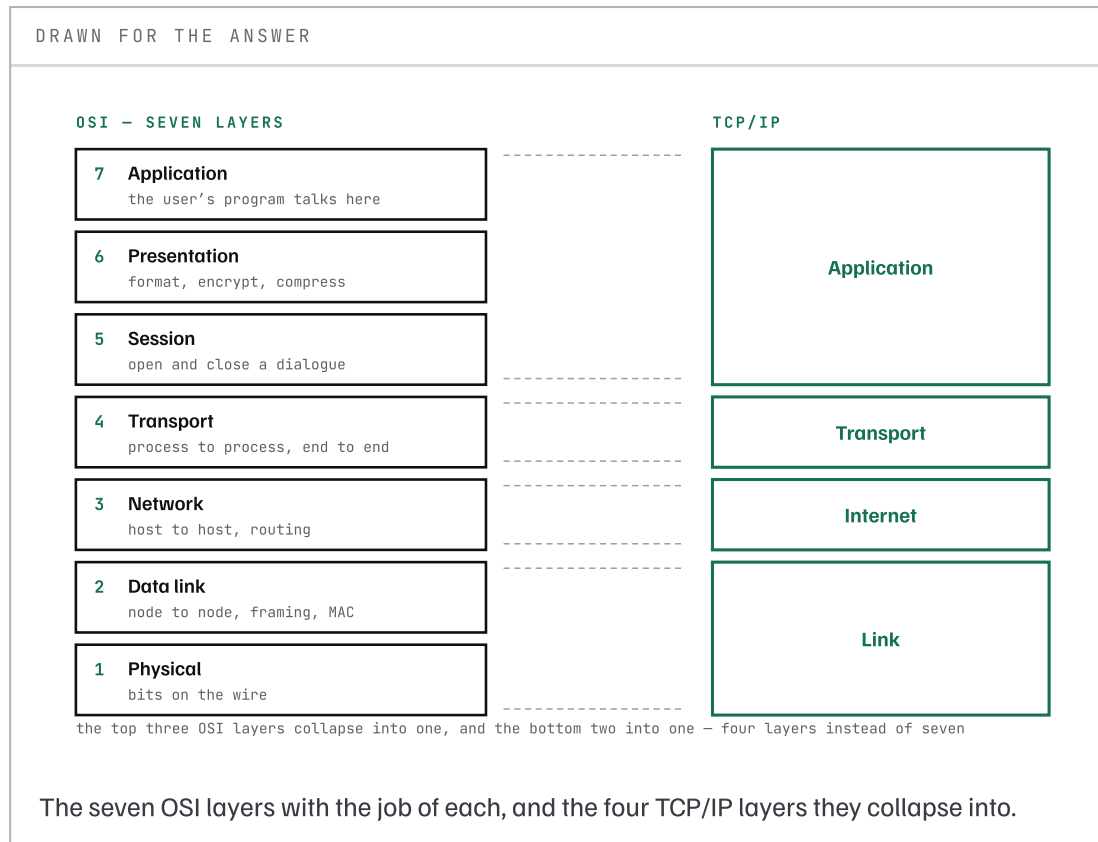
Q.1 c Write the names of each layer and the work associated with each layer.

C01

5

SOLUTION

The seven-layer OSI table from the predicted Q.2 a above. If the question says only “each layer” without naming a model, give OSI’s seven and then add the one line that TCP/IP uses five, collapsing session, presentation and application into one.



Q.1 d Draw the flow chart of the ALOHA protocol and explain it.

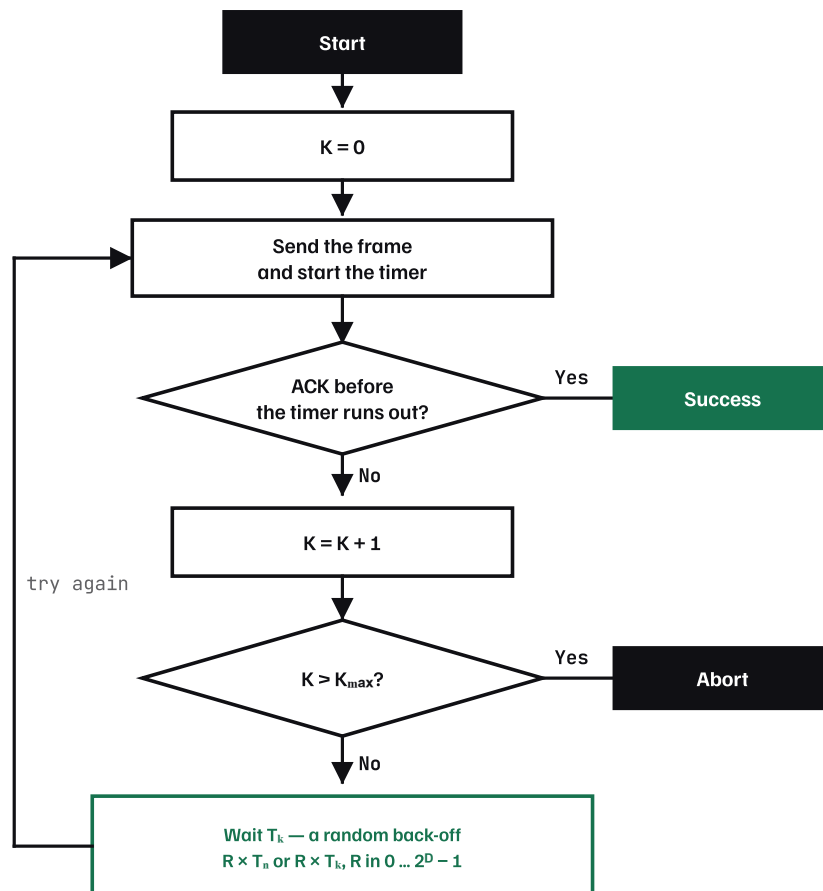
C02

3

SOLUTION

The same flow chart as 2023 and 2025. See the predicted Q.1 c.

DRAWN FOR THE ANSWER



K counts the attempts on the current frame. The wait is random on purpose – if every station waited the same time they would all collide again together.

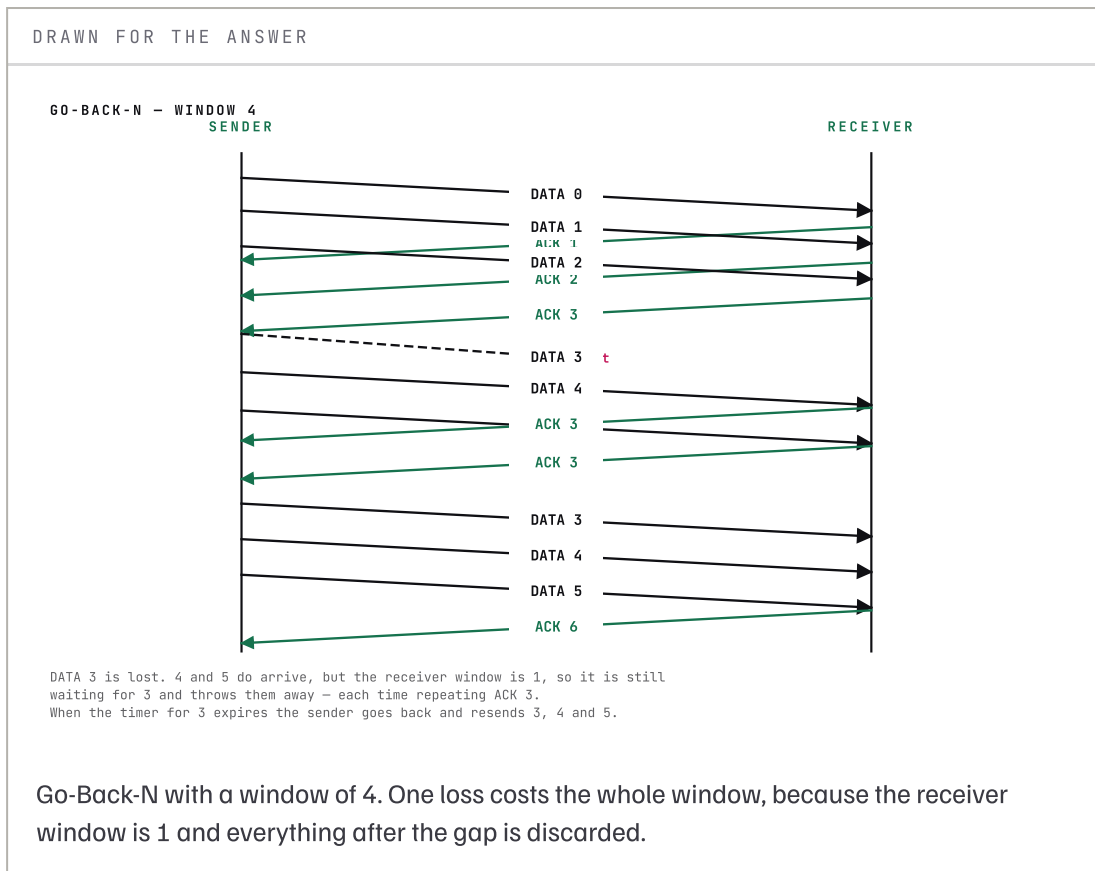
The pure ALOHA procedure. K counts the attempts; the back-off is random so that two stations that just collided do not collide again together.

Q.1 e Explain Go-back-N protocol with proper timing diagrams and sliding window diagrams. C02

3

SOLUTION

Worked in full at the 2025 Q.1 e above, with the lost-frame timing diagram and the sliding window at three stages.



Q.1 f Explain the structure of an optical fibre cable.

C01

3

SOLUTION

Structure, inside out:

- **Core** — the central glass or plastic strand the light actually travels down. It has the *higher* refractive index.
- **Cladding** — a glass layer around the core with a *lower* refractive index. The difference between the two is what traps the light.
- **Buffer / coating** — a plastic layer that protects the glass from moisture and handling damage.
- **Strength members** — usually Kevlar, so the cable can be pulled without stretching the fibre.
- **Outer jacket** — the final protective sheath.

How it guides light: total internal reflection. A ray passing from a denser medium into a less dense one bends away from the normal. Beyond a certain angle — the **critical angle** — it stops passing through at all and is reflected entirely back into

the core. So any ray entering within the acceptance cone bounces along the fibre instead of escaping.

$\theta < \text{critical angle}$ → the ray refracts out and is lost
 $\theta = \text{critical angle}$ → the ray runs along the boundary
 $\theta > \text{critical angle}$ → TOTAL INTERNAL REFLECTION, the ray stays in

Three propagation modes:

MODE	CORE	BEHAVIOUR	REACH
Single mode	Very narrow (about 8–10 μm)	Essentially one path, so almost no modal dispersion	Longest — tens of km
Multimode, step index	Wide, uniform index	Many paths of different lengths, so a pulse spreads out most	Shortest
Multimode, graded index	Wide, index falls away from the centre	The longer outer paths travel faster and partly compensate, so less spreading	Middle

Advantages: enormous bandwidth, very low attenuation, light weight, complete immunity to electromagnetic interference, and it cannot be tapped without detection. **Disadvantages:** cost, fragility, and the skill and equipment needed to splice it.

Q.2 a A sender and receiver use Selective Repeat ARQ with modulo-4 numbering and windows of size 3. Draw the timing diagram with sliding windows for this sequence: (A) frame 1 sent, received, ACK received. (B) frame 2 sent, received with an error. (C) frame 2 resent, received; ACK sent but lost. (D) frame 3 sent and received; ACK lost. (E) frame 4 sent and received; ACK lost. (F) after step E, draw the next frame the sender transmits and say whether it is correctly received. C04

5

SOLUTION

FIRST, SPOT THE TRAP

Modulo 4 means $m = 2$, so the sequence space is 4. Selective Repeat requires both windows to be at most $2^{m-1} = 2$. The question sets them to 3, which is illegal — and that is exactly the point of part (F). Say so in your answer; it is where the marks are.

SENDER	RECEIVER
A --- frame 1 ----->	accepted
<----- ACK 1 ----	ACK arrives window slides
B --- frame 2 -----X (error)	discarded / NAK
C --- frame 2 (resend) ----->	accepted
X <----- ACK 2 ----	ACK LOST
D --- frame 3 ----->	accepted
X <----- ACK 3 ----	ACK LOST
E --- frame 0 ----->	accepted (4 mod 4 = 0)
X <----- ACK 0 ----	ACK LOST
F --- frame 2 (timeout, resend) ----->	ACCEPTED AS NEW ← ERROR

Windows, step by step (sequence numbers 0,1,2,3 and back to 0):

after A : sender window {2,3,0} receiver expects {2,3,0}
 after C : sender still waiting on ACK 2 - no slide at the sender
 after E : receiver has taken 2, 3 and 0, so it now expects {1,2,3}
 the sender has had no ACKs at all since 1, so it still
 has {2,3,0} outstanding and times out on frame 2

(F) The next frame is a retransmission of frame 2 — and it is received *incorrectly*.

The receiver's window has moved on to {1, 2, 3}. Sequence number 2 is inside that window, so the receiver has no way to tell this apart from a *brand new* frame 2 from the next cycle. It accepts a duplicate as fresh data, and the data stream is silently corrupted.

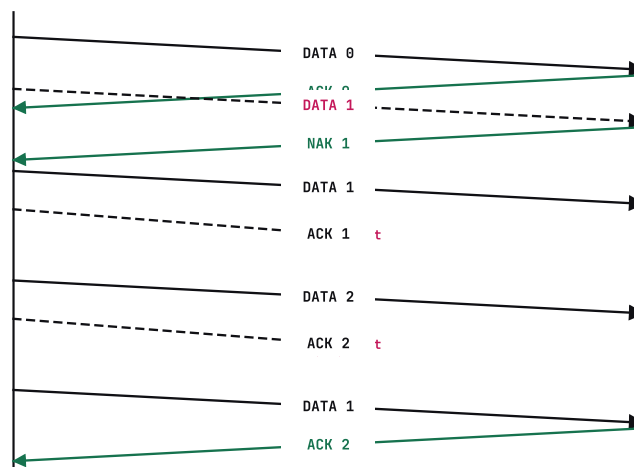
THE CONCLUSION THAT FINISHES THE ANSWER

This is precisely why Selective Repeat caps both windows at half the sequence space. With $m = 2$ the windows should be 2, not 3. Then the old and new ranges never overlap and this failure cannot happen.

DRAWN FOR THE ANSWER

SELECTIVE REPEAT - MODULO 4, WINDOWS OF 3
SENDER

RECEIVER



A: frame 0 goes and is acknowledged. B: frame 1 arrives damaged, so the receiver sends NAK 1. C: frame 1 is resent and arrives, but its ACK is lost. D: frame 2 arrives, its ACK is lost too. E: the timer for 1 expires and only frame 1 is resent – that is the whole point of Selective Repeat. F: the next frame out is DATA 3.

Selective Repeat through the six events the question lists. Only the frame that actually failed goes again.

Q.2 b

In case of CDMA, design a Walsh table for a network having 6 nodes A–F. Assign one code to each node. A transmits ‘1’, B transmits ‘0’, C transmits ‘1’, and D is receiving. Show how D detects the bits.

C01

5

SOLUTION

Word for word the same question as 2025 Q.2 b — worked in full there, using W(8) because six nodes do not fit in W(4).

DRAWN FOR THE ANSWER

WALSH W_4

A	+1	+1	+1	+1
B	+1	-1	+1	-1
C	+1	+1	-1	-1
D	+1	-1	-1	+1

A SENDS 1 · B SENDS 0 · C SENDS 1 · D LISTENS

A × (+1)

+1 +1 +1 +1

B × (-1)

-1 +1 -1 +1

C × (+1)

+1 +1 -1 -1

sum on the air

+1 +3 -1 +1

D wants A's bit, so D multiplies by A's code and averages:

(+1)(+1) + (+3)(+1) + (-1)(+1) + (+1)(+1) = 4

4 / 4 = +1 → A sent a 1

built from W_2 by $[W\ W; W\ -W]$

the codes are orthogonal, so every other station's contribution cancels to zero

The Walsh table, and one decode worked all the way through.

18 SEPTEMBER 2026

37 / 48

MID-SEM, MARCH 2023

EC322 · 30 MARKS · 15 MARCH 2023

Set for the even semester as EC322, but the same course and the same examiner style. This is the numerical-heavy paper — the only one of the three with three separate calculations.

S.V. National Institute of Technology, Surat

Department of Electronics Engineering

MID Semester Exam, March 2023

B. Tech. (EC) VI Semester · 15/03/2023

COMMUNICATION NETWORKS (EC322)

MAXIMUM MARKS: 30

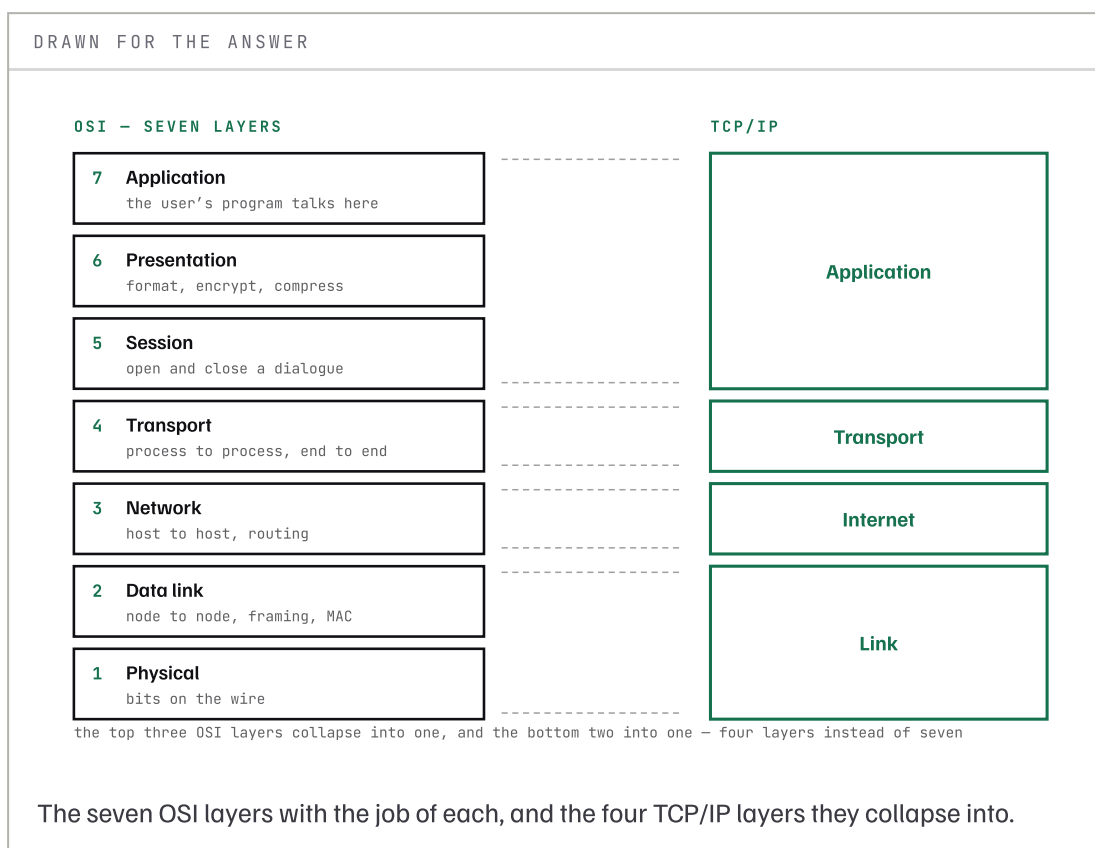
DURATION: 1:30 HOURS

ALL QUESTIONS COMPULSORY

Q.1 a Draw the TCP/IP protocol suite. Describe the work of each layer. C01

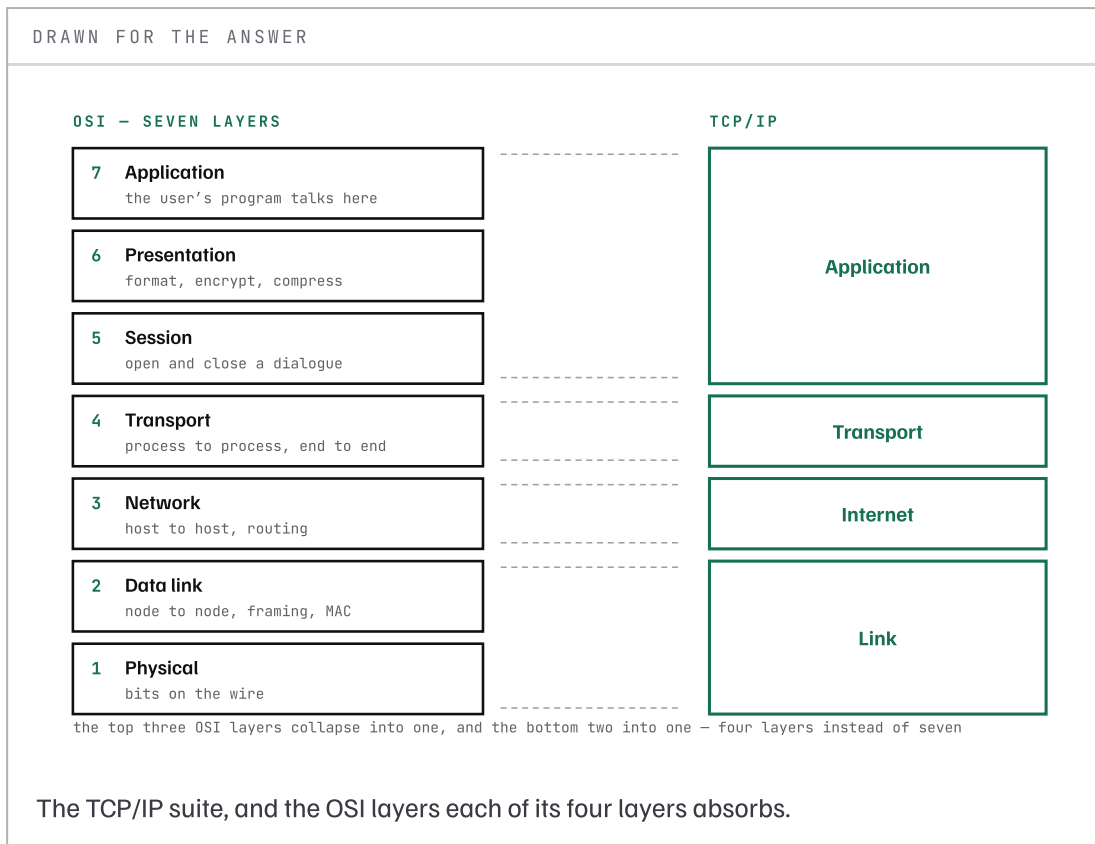
1+2

SOLUTION



- **Physical** — puts raw bits on the medium: voltages, timing, connectors.
- **Data link** — framing, physical (MAC) addressing, error and flow control, and deciding who may transmit.
- **Network** — logical (IP) addressing and routing a packet across networks.

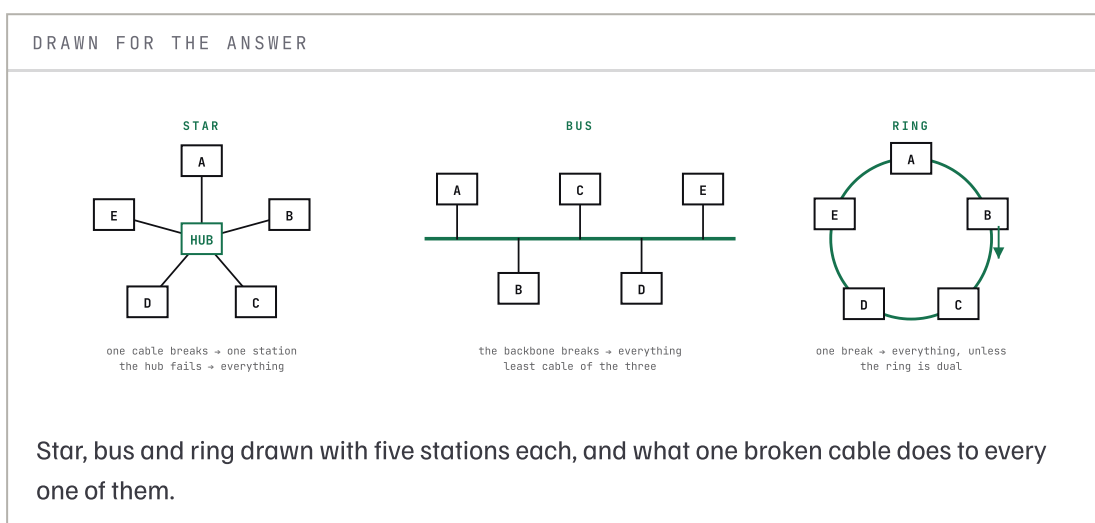
- **Transport** — process-to-process delivery using port numbers, segmentation and reassembly, end-to-end flow and error control. TCP is reliable and connection-oriented; UDP is neither and is faster for it.
- **Application** — the services the user sees. It absorbs OSI's session and presentation layers as well.



Q.1 b Take 5 nodes and connect them in mesh, bus and ring topologies. C01

3

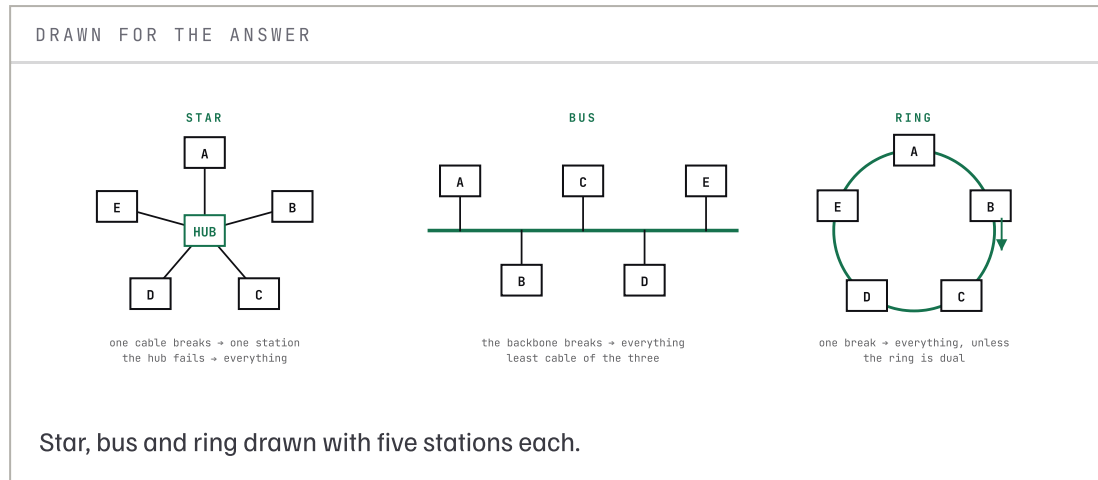
SOLUTION



Count the links to earn the mark:

mesh : $n(n-1)/2 = 5 \times 4 / 2 = 10$ links, 4 ports per node
 bus : 1 backbone + 5 drop lines
 ring : 5 links, 2 ports per node

Draw all five nodes in each case and label them. The mesh is the one people get wrong — ten links, not five.

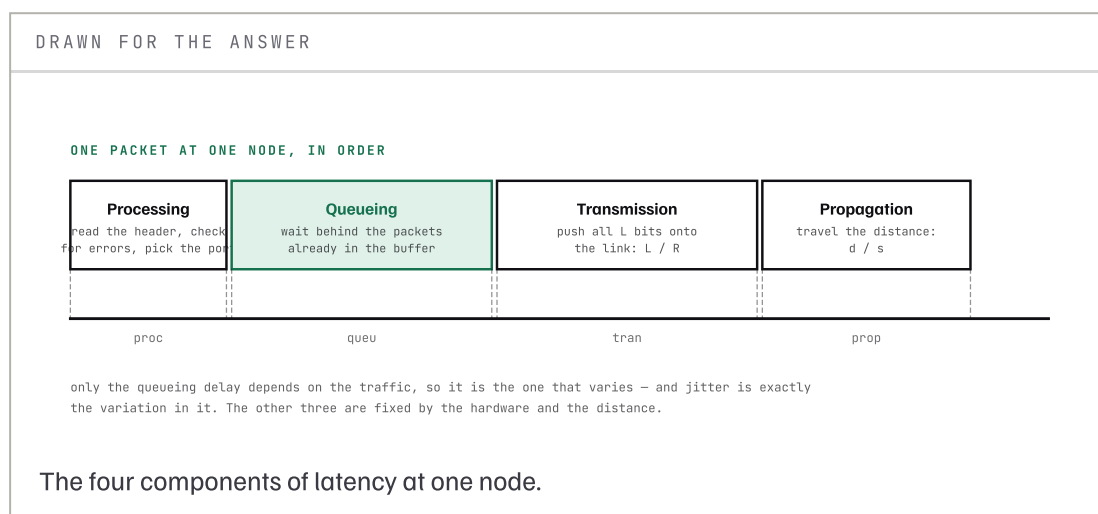


Q.1 c Explain PDR, Throughput and Latency (each component of latency). C04

3

SOLUTION

Worked in full at the predicted Q.1 d above. Note the phrase in brackets — the examiner explicitly wants all four components named, not just “delay”.

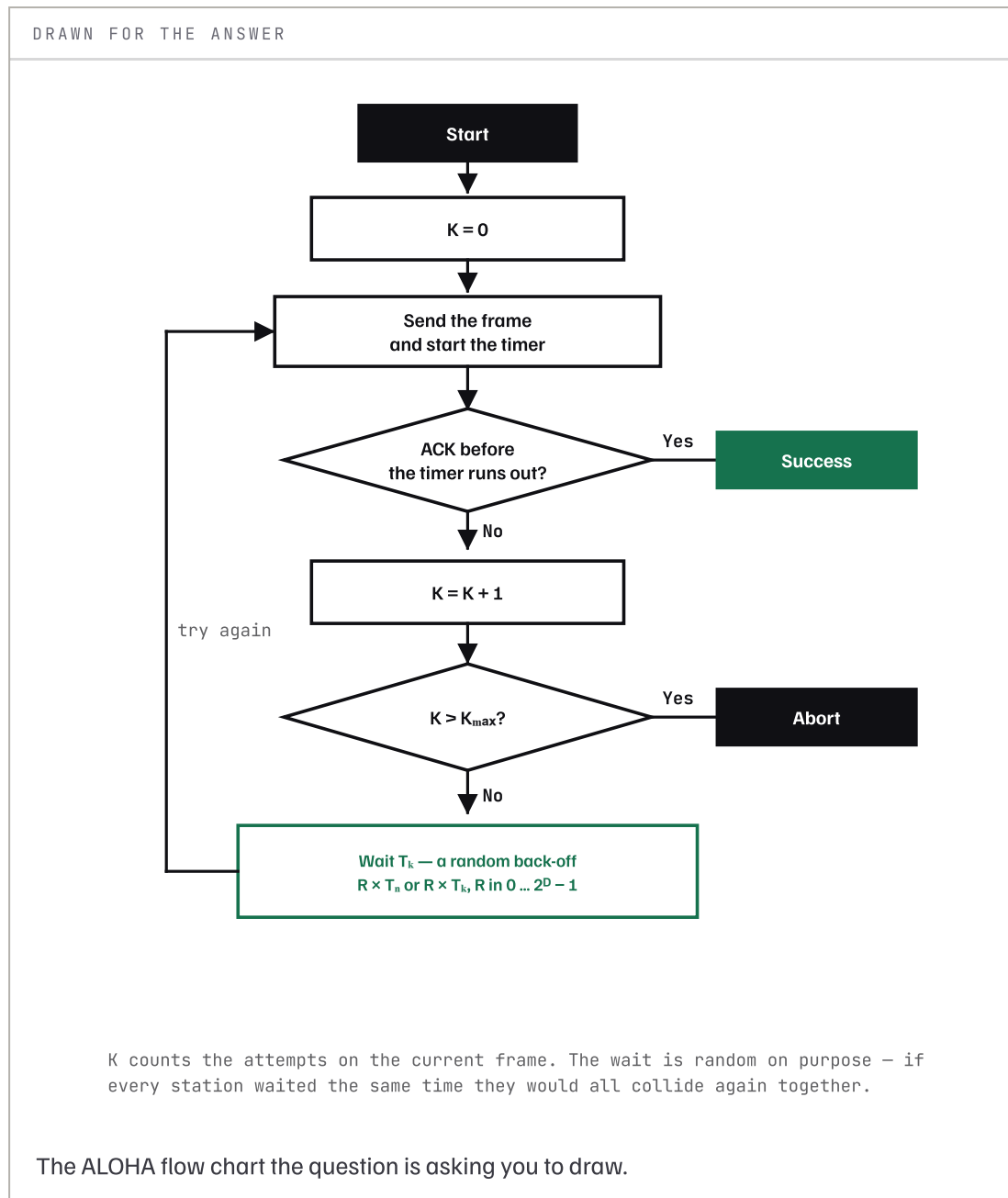


Q.1 d Draw the flow chart of ALOHA and explain each step. C02

3

SOLUTION

See the predicted Q.1 c. Same question, same answer, three years running.

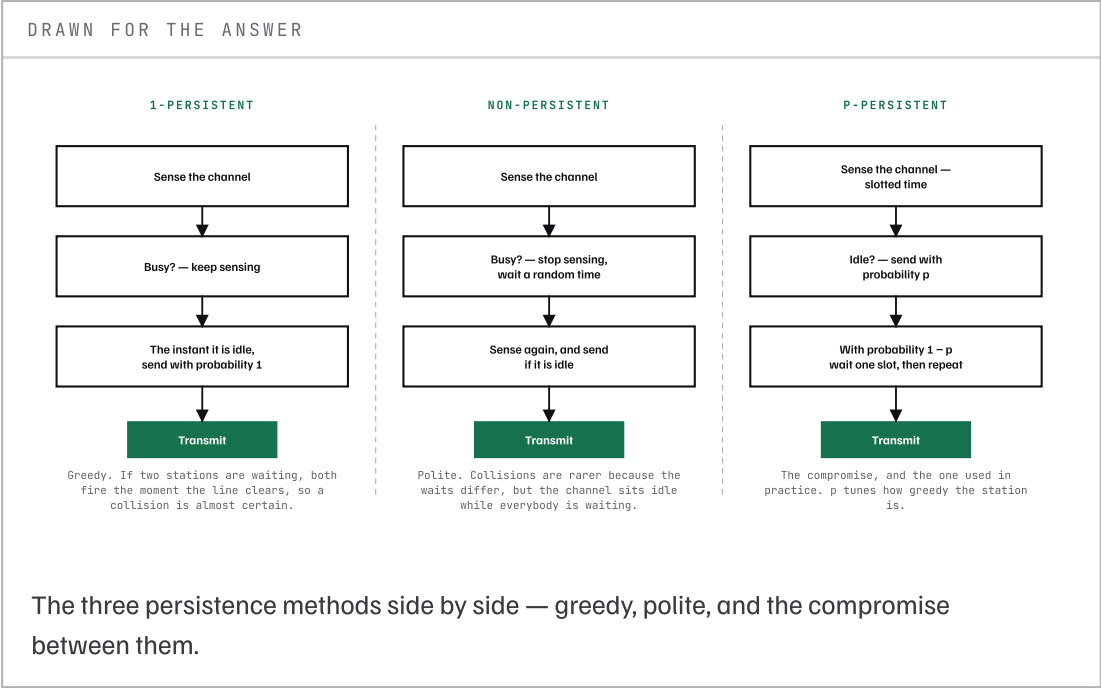


Q.1 e Draw the flow chart of the CSMA 1-persistent, non-persistent and p-persistent algorithms. C02

3

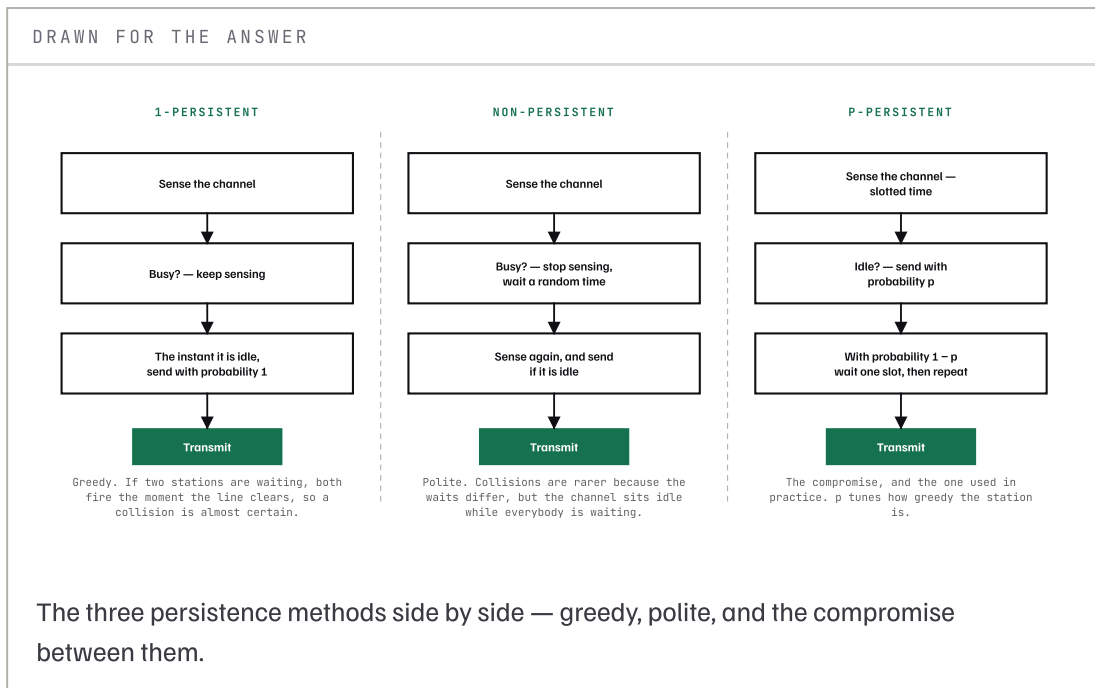
SOLUTION

All three begin by sensing the channel. They differ only in what they do when it is **busy**.



METHOD	WHEN BUSY	COLLISION RISK	CHANNEL IDLE TIME
1-persistent	Keeps sensing, sends the moment it is free	Highest — every waiting station jumps at the same instant	Lowest — the channel is grabbed immediately
Non-persistent	Waits a random time, then senses again	Lowest	Highest — the channel can sit idle while everyone is backing off
p-persistent	Sends with probability p in each idle slot	Tunable — between the two	Tunable

p-persistent is the compromise: it needs a slotted channel, and p is chosen so that the expected number of stations transmitting in a slot is about one.



Q.2 a Write the differences among the passive hub, active hub, 2-layer switch, router and gateway. C01

3

SOLUTION

DEVICE	LAYER	WHAT IT DOES	COLLISION / BROADCAST DOMAIN
Passive hub	—	Just a junction point for the wires. No power, no regeneration; the signal is only split.	One collision domain, one broadcast domain
Active hub	Physical (1)	A multiport repeater: it <i>regenerates</i> the signal before flooding it out of every other port. Still address-blind.	One collision domain, one broadcast domain
2-layer switch	Data link (2)	Reads the destination MAC and forwards the frame only to the correct port, using a table it learns from source addresses.	One collision domain per port; one broadcast domain

DEVICE	LAYER	WHAT IT DOES	COLLISION / BROADCAST DOMAIN
Router	Network (3)	Reads the destination IP, chooses a route, rebuilds the frame for the next hop and decrements the TTL.	Separate collision <i>and</i> broadcast domains
Gateway	All layers	A protocol converter — joins two networks that speak completely different protocol stacks, translating between them.	Full separation

The one-line distinction worth writing: **a hub floods, a switch filters, a router routes, and a gateway translates.**

Q.2 b A pure ALOHA network transmits 100-bit frames on a shared channel of 200 kbps. What is the throughput if the system produces (a) 1000 frames per second, (b) 200 frames per second, (c) 50 frames per second? C01

3

SOLUTION

Frame time, once for all three parts. This is also how many frame-slots a second holds:

$$T(\text{fr}) = 100 \text{ bits} / 200,000 \text{ bps} = 0.5 \text{ ms}$$

$$\text{so one second holds } 1 / 0.0005 = 2000 \text{ frame times}$$

Throughput S is measured **per frame time**, so successful frames per second = $S \times 2000$.

(a) 1000 frames per second offered

$$G = 1000 \times 0.5 \text{ ms} = 0.5$$

$$S = G e^{-(2G)} = 0.5 \times e^{-1} = 0.184$$

$$\text{successes/s} = 0.184 \times 2000 = 368$$

(a) 368 frames per second – 36.8% of the 1000 offered

(b) 200 frames per second offered

$$\begin{aligned}
 G &= 200 \times 0.5 \text{ ms} &= 0.1 \\
 S &= 0.1 \times e^{(-0.2)} = 0.1 \times 0.8187 = 0.0819 \\
 \text{successes/s} &= 0.0819 \times 2000 &= 164
 \end{aligned}$$

(b) about 164 frames per second – 81.9% of the 200 offered

(c) 50 frames per second offered

$$\begin{aligned}
 G &= 50 \times 0.5 \text{ ms} &= 0.025 \\
 S &= 0.025 \times e^{(-0.05)} &= 0.0238 \\
 \text{successes/s} &= 0.0238 \times 2000 &= 47.6
 \end{aligned}$$

(c) about 48 frames per second – 95.1% of the 50 offered

READ THE PATTERN, AND GIVE BOTH NUMBERS

As the offered load drops, the *number* getting through drops — 368, 164, 48 — but the *fraction* getting through climbs: 36.8%, 81.9%, 95.1%. Pure ALOHA is at its most efficient when it is barely used.

The absolute peak is at $G = 0.5$, where $S = 0.184$ — which is case (a). Beyond that, offering more traffic delivers *less*, because collisions grow faster than attempts.

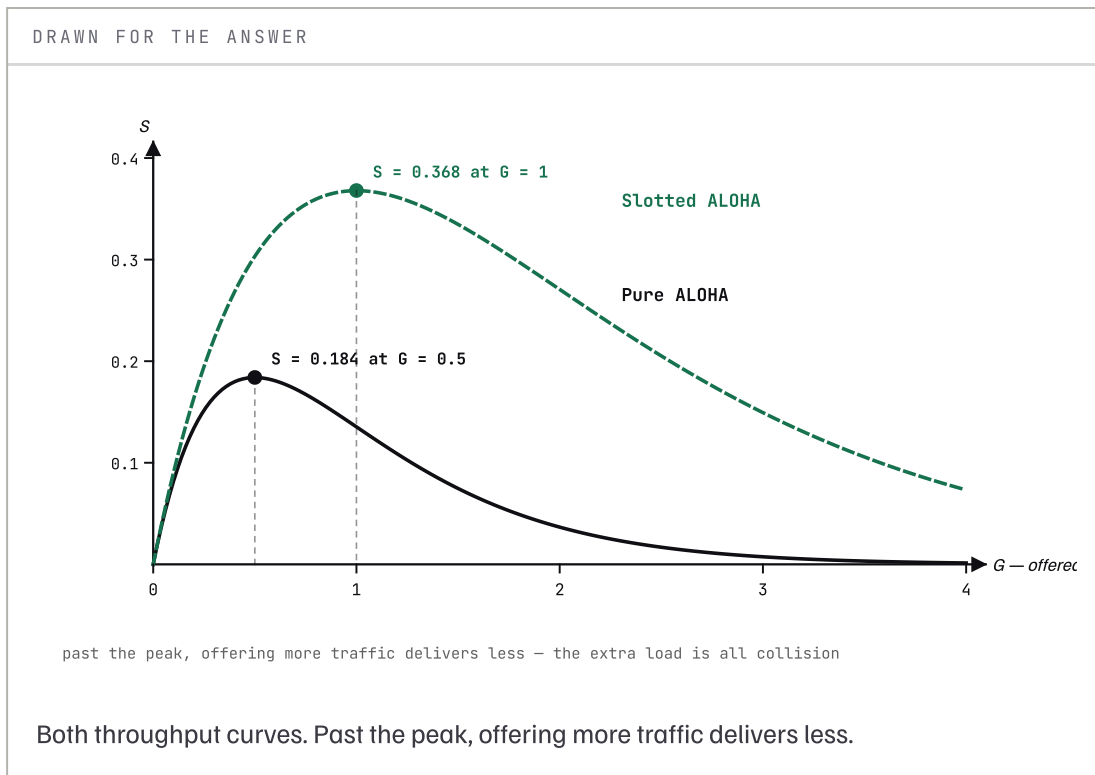
Notice the shortcut hiding in the arithmetic: the fraction succeeding is just $S/G = e^{-2G}$, so 0.368, 0.819 and 0.951 could have been written straight down.

COMPUTED · TOOLS/MAKE-FIGURES.PY



Computed figure

The throughput curves, simulated rather than quoted.



Q.3 a A network using CSMA/CD has a bandwidth of 10 Mbps. If the maximum propagation time (including device delays, ignoring the jamming signal) is 51.2 μs , what is the minimum size of the frame? C01

3

SOLUTION

$$T(\text{frame}) \geq 2 \times T(\text{propagation}) = 2 \times 51.2 \mu\text{s} = 102.4 \mu\text{s}$$

$$\text{minimum frame} = 10 \times 10^6 \times 102.4 \times 10^{-6} = 1024 \text{ bits}$$

1024 bits = 128 bytes

Note that this paper gives the **one-way** propagation time, so it must be doubled. Standard Ethernet's 64-byte minimum comes from the same rule with $T_p = 25.6 \mu\text{s}$ — half of this.

Q.3 b How can the IFS be used to prioritise a node? C02

2

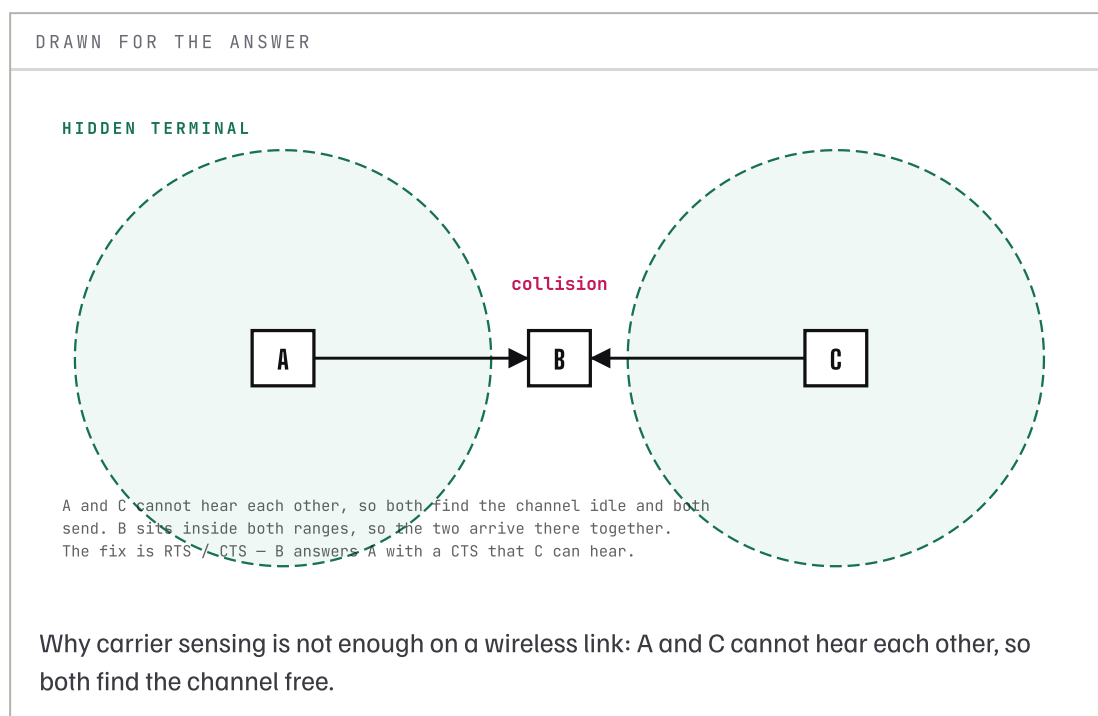
SOLUTION

IFS — interframe space. In CSMA/CA, a station that finds the channel idle does not transmit at once. It first waits a fixed interframe space, and only then starts its back-off countdown.

The priority mechanism: give different classes of traffic **different IFS lengths**. A station with a **shorter IFS** finishes waiting first, so it starts — and usually wins — before a station with a longer IFS has even begun. The longer-IFS station then senses the channel busy and defers.

SIFS (shortest)	- acknowledgements, CTS: highest priority
PIFS	- point coordination, contention-free traffic
DIFS (longest)	- ordinary data: lowest priority

So the rule is simply: **shorter IFS means higher priority**. It also has a second job — it stops two stations from transmitting immediately after the channel clears, which would otherwise guarantee a collision.



Q.4 a In a Stop-and-Wait ARQ system the bandwidth of the line is 1 Mbps and 1 bit takes 20 ms to make a round trip. What is the bandwidth-delay product? If the data frames are 1000 bits long, what is the utilisation percentage of the link?

2

SOLUTION

$$\text{BDP} = 1 \times 10^6 \times 20 \times 10^{-3} = 20,000 \text{ bits}$$

$$\text{utilisation} = 1000 / 20,000 = 0.05$$

$$\text{BDP} = 20,000 \text{ bits} \cdot \text{utilisation} = 5\%$$

The pipe holds 20,000 bits, and Stop-and-Wait only ever puts 1000 of them in flight at a time. The link is idle 95% of the time, waiting for an acknowledgement — which is the entire argument for sliding-window protocols.

Q.4 b What is the utilisation percentage of the link in 4(a) if we use a protocol that can send up to 15 frames before stopping to worry about acknowledgements? C01

2

SOLUTION

$$\begin{aligned} \text{bits in flight} &= 15 \times 1000 = 15,000 \\ \text{utilisation} &= 15,000 / 20,000 = 0.75 \end{aligned}$$

75%

To reach 100% the window would need to be $20,000 / 1000 = 20$ frames. The general rule: the window must be at least the bandwidth-delay product divided by the frame size.